

Age-related changes in “cortical” 1/f dynamics are linked to cardiac activity

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Examination of (a)periodic brain activity has gained particular interest in the last few years in the neuroscience fields relating to cognition, disorders, and brain states. Using large EEG/MEG datasets from younger and older adults, the current study provides **compelling** evidence that age-related differences in aperiodic EEG/MEG signals can be driven by cardiac rather than brain activity. Their findings have **important** implications for all future research that aims to assess aperiodic neural activity, suggesting control for the influence of cardiac signals is essential.

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Abstract

The power of electrophysiologically measured cortical activity decays with an approximately $1/f^X$ function. The slope of this decay (i.e. the spectral exponent, X) is modulated by various factors such as age, cognitive states or psychiatric/neurological disorders. Interestingly, a mostly parallel line of research has also uncovered similar effects for the spectral slope in the electrocardiogram (ECG). This raises the question whether these bodywide changes in spectral slopes are (in-)dependent. Focusing on well-established age-related changes in spectral slopes we analyzed a total of 1282 recordings of magnetoencephalography (MEG) resting state measurements with concurrent ECG in an age-diverse sample (18-88 years). Using a diverse array of analytical approaches, we demonstrate that the aperiodic signal recorded via surface electrodes/sensors originates from multiple physiological sources. Furthermore, our results suggest that common “artifact” rejection approaches (i.e. ICA) may not be sufficient to separate cardiac from neural activity. In particular, significant parts of age-related changes in aperiodic activity normally interpreted to be of neural origin can be explained by cardiac activity. Moreover, our results suggest that changes (flattening/steepening) of the spectral slope with age are dependent on the recording site and investigated frequency range. Our results highlight the complexity of aperiodic activity while raising concerns when interpreting aperiodic activity as “cortical” without considering physiological influences.

Introduction

Aperiodic neural activity is omnipresent both in invasive (e.g. ECoG¹) and non-invasive (e.g. MEG/EEG^{2,3}) recordings of electrophysiological brain activity and even in hemodynamic responses (e.g. fMRI⁴). In the frequency domain, when visualized in log-log coordinates (log-frequency/log-power), aperiodic activity manifests as a linear decay in power with an increase in frequency¹ (i.e. the spectral slope). This part of the signal - following a so-called “power-law” distribution - is often referred to as “scale-free”, “1/f noise” or more recently “aperiodic activity”⁵ (see **Figure 1A** for an illustration of aperiodic activity with different spectral slopes in the time and frequency domain).

In the past, aperiodic neural activity was often treated as noise and simply removed from the signal e.g. via pre-whitening^{6,7}, so that analyses could focus on periodic neural activity (local peaks that rise above the “power-law” distribution, which are commonly thought to reflect neural oscillations). However, in recent years the analysis of aperiodic neural activity has gained interest (see **Figure 1D**). Several studies have shown that aperiodic neural activity is meaningfully modulated by various factors, such as age⁸, cognitive state (e.g. awake vs. sleep⁴) and disorders like Parkinson’s disease and epilepsy^{9,10}. However, aperiodic activity is not only present in recordings of neural activity, but also part of other physiological signals such as cardiac and muscle activity, commonly measured using electrocardiography (ECG¹¹) and electromyography¹². Interestingly, and mostly overlooked by the neuroscience community (see **Figure 1C**), aperiodic activity measured using ECG (often referred to as power law or 1/f activity) is modulated by similar factors as neural aperiodic activity, including aging¹³, different cognitive states (eg. awake vs. sleep^{12,14}) and disorders such as Parkinson’s disease and epilepsy^{15,16} (see also **Figure 1B**).

Furthermore, it is well-known that, via volume conduction, cardiac activity can also be captured in both invasive and non-invasive recordings of neural activity^{17–19}. Hence, it is also considered best practice to measure and remove cardiac activity from M/EEG recordings²⁰. However, an analysis of openly accessible M/EEG articles that investigate aperiodic activity ($N_{Articles}=279$; see Methods - Literature Analysis for further details) revealed that only 17.1% of EEG studies explicitly mention that cardiac activity was removed and only 16.5% measure ECG (45.9% of MEG studies removed cardiac activity and 31.1% of MEG studies mention that ECG was measured; see **Figure 1EF**). Additionally, investigations of aperiodic activity vary strongly by both the upper and lower bounds and the general width of the analyzed frequency ranges (1GHI). This further complicates the comparison of results across studies as physiological signals (e.g., cardiac activity) may have varying impacts across frequency ranges—for instance, exerting a stronger influence on lower frequencies than on higher ones.

Considering that A) aperiodic neural and cardiac activity are modulated by similar traits, states, and disorders, and B) cardiac activity is often present but rarely removed from neural recordings, we ask: Are changes in aperiodic neural activity (in-)dependent from changes in aperiodic cardiac activity? To address this question, we turn our attention to the recently reported^{5,8} and replicated²⁷ association between aperiodic activity and chronological age. Using the publicly available Cam-CAN dataset^{28,29}, we find that the aperiodic signal measured using M/EEG originates from multiple physiological sources. In particular, significant portions of age-related changes in aperiodic activity –normally attributed to neural processes– can be better explained by cardiac activity. This observation holds across a wide range of processing options and control analyses (see **Supplementary S1**), and was replicable on a separate MEG dataset. However, the extent to which cardiac activity accounts for age-related changes in aperiodic activity varies with the investigated frequency range and recording site. Importantly, in some frequency ranges and sensor locations, age-related changes in neural aperiodic activity still prevail. But does the

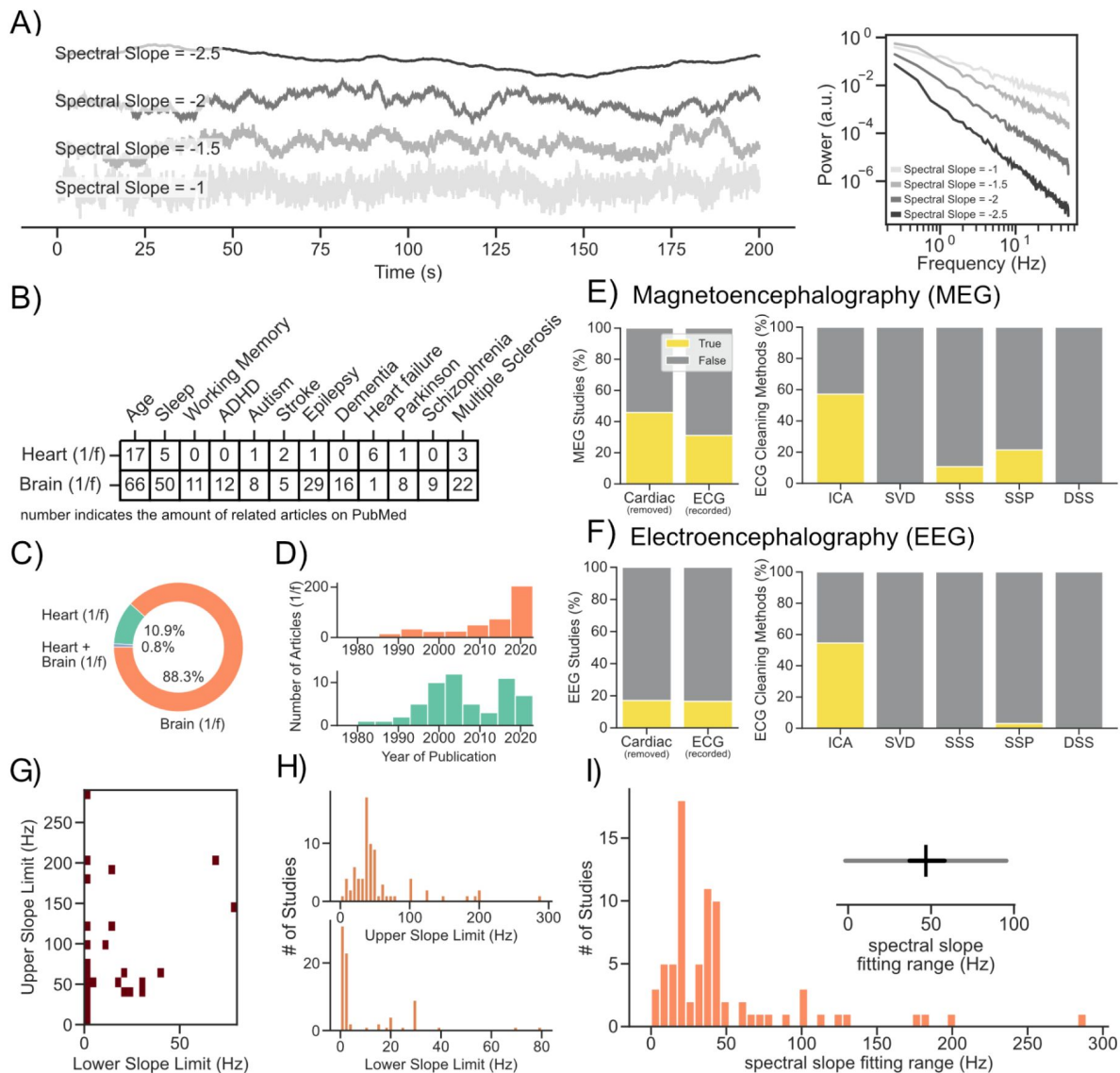


Figure 1

Literature analysis of aperiodic activity investigated using M/EEG and ECG.

A) Illustration of different types of aperiodic activity in the time and frequency domain. **BC)** We analyzed 489 abstracts indexed on PubMed using LISC²¹, a package for collecting and analyzing scientific literature. **B)** This analysis revealed that changes in aperiodic activity are related to similar traits, states and disorders in measures of both neural and cardiac activity. **C)** We further noted a tiny overlap of studies (N=4) that refer to both cardiac and cortical aperiodic activity in their abstracts. Yet, none of these studies considers confounding influences of cardiac aperiodic activity on the measurement of cortical aperiodic activity. **D)** We additionally found a steep increase related to the investigation of neural aperiodic activity in the 2020s highlighting the current interest of the topic in the neuroscience community. **EF)** We further downloaded and analyzed freely available full-texts of M/EEG studies investigating aperiodic activity to see to which extent and how cardiac activity was handled. This analysis revealed that only 17.1% of EEG studies remove cardiac activity and only 16.5% measure ECG (for MEG 45.9% removed cardiac; 31.1% mention ECG was measured). We were further interested in determining which artifact rejection approaches were most commonly used to remove cardiac activity, such as independent component analysis (ICA²²), singular value decomposition (SVD²³), signal space separation (SSS²⁴), signal space projections (SSP²⁵) and denoising source separation (DSS²⁶). We found that the most commonly applied method both in EEG and MEG recordings was independent component analysis (ICA). **GH)** An arbitrary selection of previous studies (N = 60) shows a vast amount of different frequency ranges are used to investigate aperiodic activity. While a significant amount of studies looked into a range between ~0.1-50 Hz (~30%), most studies used a unique frequency range. **I)** Not only do the upper and lower bounds vary between studies, but also the general width of the fitting range can vary from 0.9 to 290 Hz.

influence of cardiac activity on the aperiodic spectrum extend beyond age? In a preliminary analysis, we demonstrate that working memory load modulates the aperiodic spectrum of “pure” ECG recordings. The direction of this working memory effect mirrors previous findings on EEG data⁵, suggesting that the impact of cardiac activity goes well beyond aging. In sum, our results highlight the complexity of aperiodic activity while cautioning against interpreting it as solely “neural” without considering physiological influences.

Results

Aperiodic signals recorded using ECG are associated with aging and heart rate variability

Changes of aperiodic activity in recordings of neural and cardiac activity are associated with aging^{8,13}. However, analyses of ECG signals - in the frequency domain - typically focus on (a-)periodic signals <0.4Hz³⁰. These (compared to neural time series) slowly fluctuating signals are related to heart rate variability³¹. Changes in (a-)periodic activity at these low frequencies are well established physiological measures³². However, substantially less is known about aperiodic activity above 0.4Hz in the ECG. Yet, common ECG setups for adults capture activity at a broad bandwidth of 0.05 - 150Hz^{33,34}. Importantly, a lot of the physiological meaningful spectral information rests between 1-50Hz³⁵, similarly to M/EEG recordings. Furthermore, meaningful information can be extracted at much higher frequencies. For instance, ventricular late potentials have a broader frequency band (~40-250Hz³⁵). However, that’s not all, as further meaningful information can be extracted at even higher frequencies (>100Hz). For instance, the so-called high-frequency QRS seems to be highly informative for the early detection of myocardial ischemia and other cardiac abnormalities that may not yet be evident in the standard frequency range^{36,37}. Yet, the exact physiological mechanisms underlying the high-frequency QRS remain unclear (see ³⁷ for a review discussing possible mechanisms).

To understand whether aperiodic activity recorded using ECG carries meaningful information about aging –at frequency ranges common in M/EEG recordings– the ECG data of 4 age-diverse populations with a total of 2286 subjects were analyzed.

After pre-processing (see Methods), age was used to predict the spectral slope of the ECG over various different frequency ranges (see **Figure 2C**). Due to the presence of a “knee” in the ECG data (for details regarding “knees” in power spectra see^{1,5,38}), slopes were fitted individually to each subject’s power spectrum in several lower (0.25 – 20 Hz) and a higher (10-145 Hz) frequency ranges. The split in lower and higher frequency ranges was performed to avoid spectral knees at ~15 Hz in the center of the slope fitting range (see **Supplementary Figure 1B** for the distribution of knee frequencies across datasets). Our results show that the spectral slope flattened with age over a vast amount of different frequency ranges (see **Figure 2C**). These results are similar to what was reported in previous studies measuring “cortical” aperiodic activity. However, we also noted an age-related steepening of the spectral slope in one dataset (ECG Dataset 3 - Cam-Can) in the low frequency range (0.25 – 12 Hz, see Discussion).

But do these aperiodic changes at the ECG also correspond to established indices of cardiac health and function? To better understand this, we conducted an exploratory analysis, where we related the spectral slope of the aperiodic ECG signal in various frequency ranges to 90 different indices of heart rate variability (implemented in NeuroKit2³⁹) across all 4 datasets. The results show that spectral slopes both in lower (0.25 - 20 Hz) and higher (10 - 145 Hz) frequency ranges relate to several indices of heart rate variability (see **Figure 2D**). Overall, spectral slopes in lower frequency ranges were more consistently related to heart rate variability indices (39.4-74.6% percent of all investigated indices) than spectral slopes in higher frequency ranges (16-30.01% percent of all investigated indices; see **Figure 2D**). In the lower frequency ranges (0.25 - 20Hz),

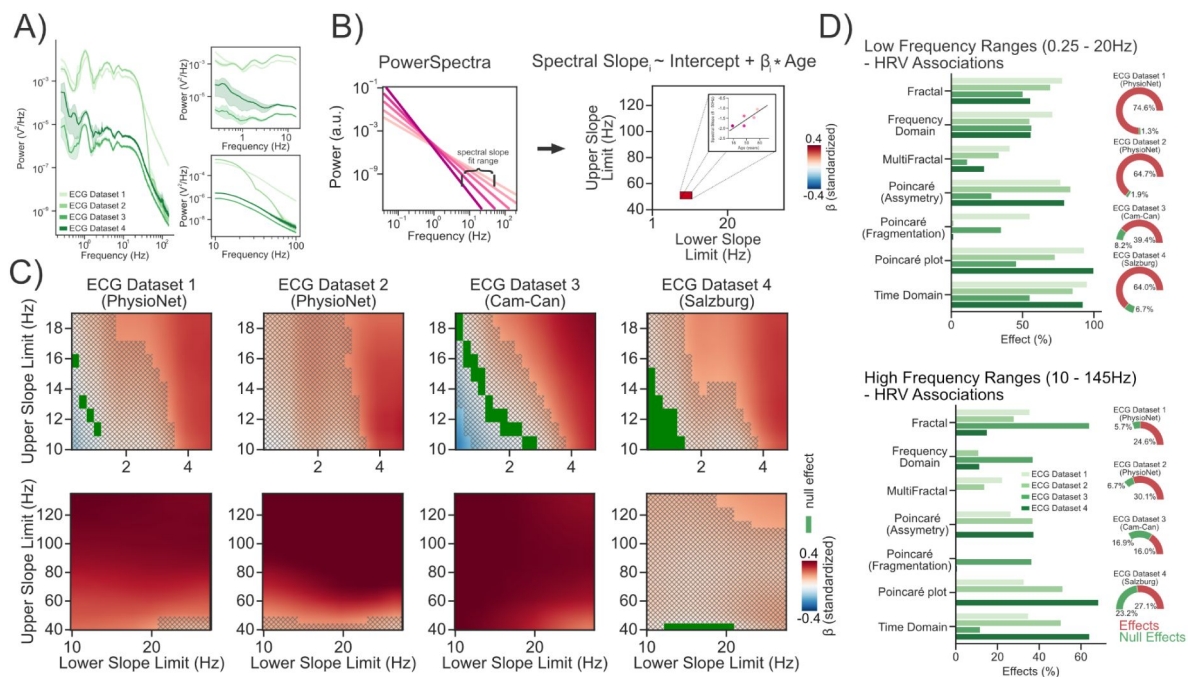


Figure 2

Aperiodic signals recorded using ECG are related to aging and heart rate variability.

A) grand average power spectra plotted separately per Dataset and the associated aperiodic power spectra for the lower (0.25-20 Hz) and higher (10-145 Hz) frequency ranges. **BC)** Age was used to predict the spectral slope using different upper and lower slope limits for higher (10 - 145 Hz) and lower (0.25 - 20Hz) frequency ranges. Significant effects, i.e. effects with credible intervals not overlapping with a region of practical equivalence (ROPE; see Methods - Statistical Inference), are highlighted in red or blue (see colorbar). Null effects, which were defined as effects with credible intervals completely within a ROPE, are highlighted in green. Results where no decision to accept or reject (see⁴⁰) an effect could be made, are masked using hatches. **D)** To understand whether aperiodic cardiac activity also relates to common measures of heart rate variability we predicted the spectral slope using 90 different measures of heart rate variability. We find consistent (yet different) associations with mostly fractal and time domain measures in both lower and higher frequency ranges.

spectral slopes were consistently related to most measures of heart rate variability; i.e. significant effects were detected in all 4 datasets (see **Figure 2D** [↗](#)). This includes fractal, multifractal, time and frequency domain analyses as well as indices extracted from the Poincaré plot. In the higher frequency ranges (10 - 145 Hz) spectral slopes were most consistently related to fractal and time domain indices of heart rate variability, but not so much to frequency-domain indices assessing spectral power in frequency ranges < 0.4 Hz. This suggests that spectral slopes > 10 Hz carry meaningful information about cardiac activity that is largely distinct from the frequency-domain information that is commonly investigated using ECG. In sum, these findings show that aperiodic activity, in frequency ranges that vastly exceeds those commonly explored in ECG analyses, may carry meaningful information about cardiac activity.

With regards to aging, the conducted analyses show that aperiodic activity measured via ECG is associated with aging at frequency ranges vastly exceeding those typically investigated via ECG, but overlapping with frequency ranges commonly measured in recordings of neural activity (see **Figure 1G** [↗](#)). Importantly, the direction of the association between age and aperiodic ECG activity is largely identical to that reported for age and aperiodic EEG activity^{5,8} [↗](#), motivating the investigation of these relationships in combined neural and cardiac measurements.

Cardiac activity is directly captured in EEG and MEG recordings

Aperiodic activity recorded using ECG (see **Figure 2C** [↗](#)) and EEG/ECOG⁸ [↗](#) is similarly modulated by age. In MEG and some EEG recordings cardiac activity is measured via ECG²⁰ [↗](#). Components of the signal related to cardiac activity are then commonly removed via independent component analysis (ICA²² [↗](#); see **Figure 1EF** [↗](#)). In recordings of EEG the influence of cardiac activity is often deemed less problematic¹⁷ [↗](#), as a result ECG is rarely recorded (see **Figure 1E** [↗](#)).

We utilized concurrent ECG, EEG and MEG resting state recordings to examine to what extent ECG signals are present in the signals measured using MEG and EEG. We calculated so-called temporal response functions (see Methods), to detect whether the signals recorded at different locations (M/EEG vs. ECG) are instantaneously related (zero-time-lags; and therefore likely correspond to the same source) or if one lags behind the other (non zero-time-lags; likely different sources influencing each other e.g. via interoception see⁴¹ [↗](#)). After pre-processing (see Methods) the data was split in three conditions using an ICA²² [↗](#). Independent components that were correlated (at $r > 0.4$; see Methods: *MEG/EEG Processing - pre-processing*) with the ECG electrode were either not removed from the data (**Figure 3ABCD** [↗](#) - blue), removed from the data (**Figure 2ABCD** [↗](#) - orange) or projected back into the sensor space (**Figure 3ABCD** [↗](#) - green). Afterwards, temporal response functions (encoding models; see Methods) between the signal recorded at the ECG electrode and the MEG/EEG sensors and feature reconstruction models (decoding models) were computed (for each condition respectively). Our results show that if ECG components are not removed via ICA, the ECG signal is captured equally strong at zero-time-lags both in EEG and MEG recordings (see **Figure 3ACD** [↗](#)). Even after removing ECG related components from the data TRF peaks emerged (although reduced) at zero-time-lags in EEG, but not in MEG recordings (see **Figure 3AB** [↗](#)). Furthermore, reconstruction (decoding) of the ECG signal was reduced, but remained above chance even after rejecting the ICA signal using ECG both in MEG and EEG recordings ($r > 0$). Interestingly, the presence of the ECG signal was more pronounced in EEG compared to MEG recordings, after removing ECG related components ($\beta_{\text{standardized (EEG > MEG)}} = 0.97$, $HDI = [0.42, 1.52]$; **Figure 3D** [↗](#)). Additionally, ECG related components extracted from MEG recordings were more related to the ECG signal than the components extracted from the EEG ($\beta_{\text{standardized (EEG > MEG)}} = -0.76$, $HDI = [-1.35, -0.18]$; **Figure 3D** [↗](#)). These results show that A) residual ECG activity remains in surface neural recordings, even after applying a very sensitive threshold to detect and remove ECG components via ICA and B) neural and cardiac activity are more difficult to separate in EEG as opposed to MEG recordings (see **Figure 3ACD** [↗](#)) resulting in more residual (after ICA) ECG related activity in EEG recordings.

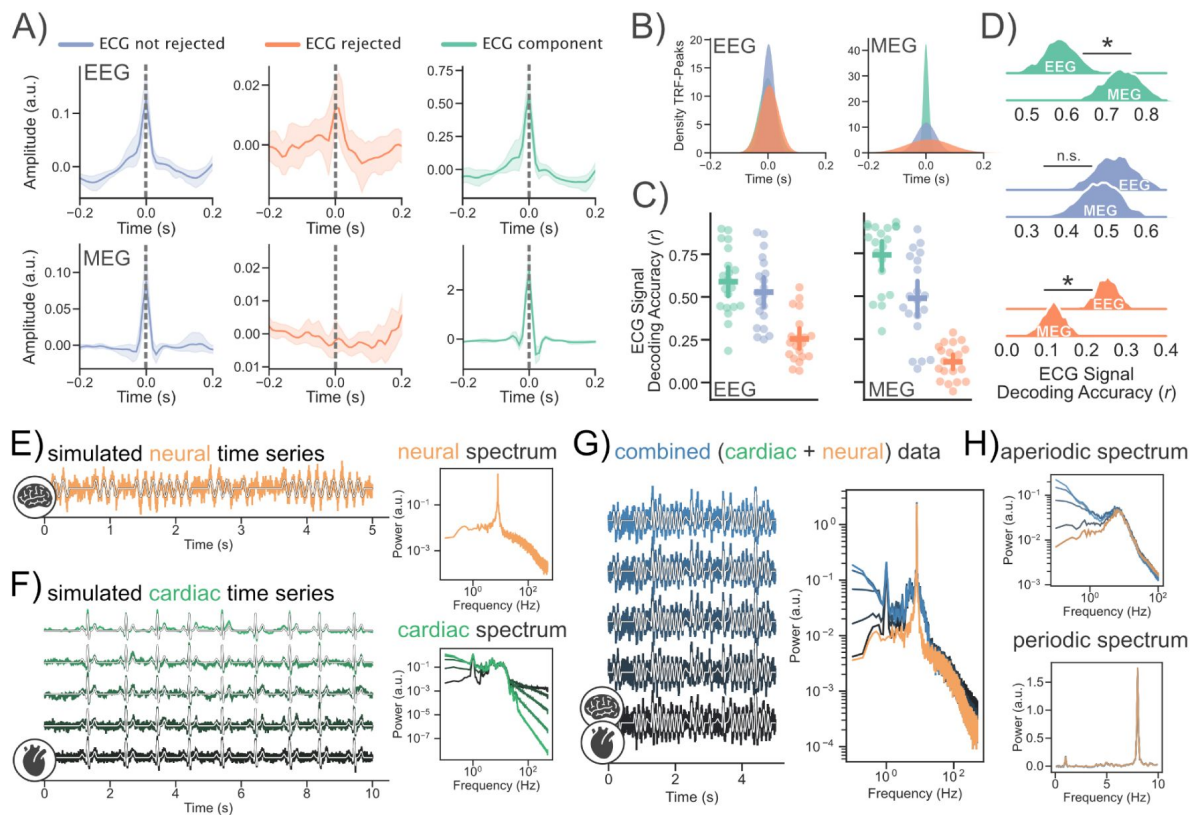


Figure 3

Cardiac activity is captured in EEG and MEG recordings.

AB) Cardiac activity is captured at zero time lags in concurrent MEG and EEG recordings, if the ECG signal is rejected via ICA this effect disappears in MEG, but not completely in EEG data. **CD)** Reconstruction of the ECG signal was impaired, but remained possible even after rejecting the ICA signal using ECG (both in MEG and EEG data). Notably, reconstruction of the ECG signal (after ICA) worked better in EEG than MEG data. A * indicates a “significant” effect (see Methods - Statistical Inference). **E)** To illustrate how aperiodic activity recorded using ECG might impact neural aperiodic activity we simulated cardiac and neural time series data. The neural time series data was simulated as in Gao et al.⁴² with an EI ratio of 1:2. The cardiac time series consists of a PQRST-Complex and different types of 1/f noise. **G)** Combining both cardiac and neural time series data shows that even if the PQRST-Complex is barely visible in the combined time domain signal, the resulting power spectrum can be heavily affected by simulated changes in aperiodic cardiac activity (**GH**).

To further illustrate how changes in aperiodic cardiac activity might impact “cortical” aperiodic activity recorded via M/EEG we simulated cardiac and neural time series data (see [Figure 3E](#)). The neural time series data was simulated as in Gao et al.⁴² with an EI ratio of 1:2. The cardiac time series consists of a simulated template PQRS-Complex at a rate of ~1Hz (with jittered onsets) and different types of additional aperiodic activity. Combining both cardiac and neural time series data shows that even if the PQRS-Complex is barely visible in the combined time domain signal, the resulting power spectrum can be heavily affected by simulated changes in aperiodic cardiac activity (see [Figure 3GH](#)).

Age-related changes in aperiodic brain activity are most pronounced in cardiac components

ECG signals are captured in brain activity recorded using M/EEG (see [Figure 3ABCD](#)). Furthermore, aperiodic activity recorded using ECG is –just like aperiodic activity recorded using EEG/ECOG– modulated by age (see [Figure 2C](#)). However, it is unclear whether these bodywide changes in aperiodic activity are (in-)dependent.

To answer this question we are leveraging resting state MEG recordings of an age-diverse population obtained from the Cam-CAN inventory ($N = 627^{28,29}$). After pre-processing (see Methods) an ICA was applied to separate MEG activity from activity related to the ECG. ICA components that were related to the ECG signal were identified using a correlation threshold ($r > 0.4$; same threshold as in [3ABCD](#)). The data was split into three conditions (MEG_{ECG} not rejected, MEG_{ECG} rejected and MEG_{ECG} component; see [Figure 3A](#)) and projected back to the sensor space respectively. Age was then used to predict the spectral slope across 102 magnetometers and over a wide variety of frequency ranges with lower limits starting at 0.5 Hz in 1 Hz steps ranging until 10 Hz and upper limits starting at 45 Hz in 5 Hz steps ranging until 145 Hz (see [Figure 4B](#)) per subject (split by condition).

This analysis, which is depicted in [Figure 4](#), shows that over a broad amount of individual fitting ranges and sensors, aging resulted in a steepening of spectral slopes across conditions (see [Figure 4B](#)) with “steepening effects” observed in 25% of the slope fits in MEG_{ECG} not rejected, 0.5% in MEG_{ECG} rejected, and 60% for MEG_{ECG} components. The second largest category of effects were “null effects” in 13% of the options for MEG_{ECG} not rejected, 30% in MEG_{ECG} rejected, and 7% for MEG_{ECG} components. However, we also found “flattening effects” in the spectral slope for 0.16% of the processing options in MEG_{ECG} not rejected, 0.3% in MEG_{ECG} rejected, and 0.46% in MEG_{ECG} components.

Interestingly, this analysis shows that over all options both flattening and steepening effects were most frequently observed on the MEG_{ECG} components. This analysis also indicates that a vast majority of observed effects irrespective of condition (ECG components, ECG not rejected, ECG rejected) show a steepening of the spectral slope with age across sensors and frequency ranges. This finding is contrary to previous findings showing a flattening of spectral slopes with age in recordings of both brain⁸ and cardiac activity¹³ (see also [Figure 2C](#)). We therefore conducted several control analyses both on data averaged across sensors (see [Supplementary Figure S2](#), S3, S4, S5, S6, S7) and on the level of single sensors (see [Supplementary Figure S8](#) & S9) to investigate to what degree this observation is based on decisions made during preprocessing (see [Supplementary Text S1 - Control Analyses: Age-related steepening of the spectral slope in the MEG](#)). In sum, all performed control analyses indicate that aging can robustly cause a steepening of the spectral slope in “cortical” activity recorded using MEG that is not explainable by age-related changes in head-movements or EOG activity, the application of different blind source separation algorithms (e.g. ICA and SSS), the algorithm used to extract the spectral slope (IRASA vs. FOOOF), and replicable across two large MEG datasets. These steepening effects have previously

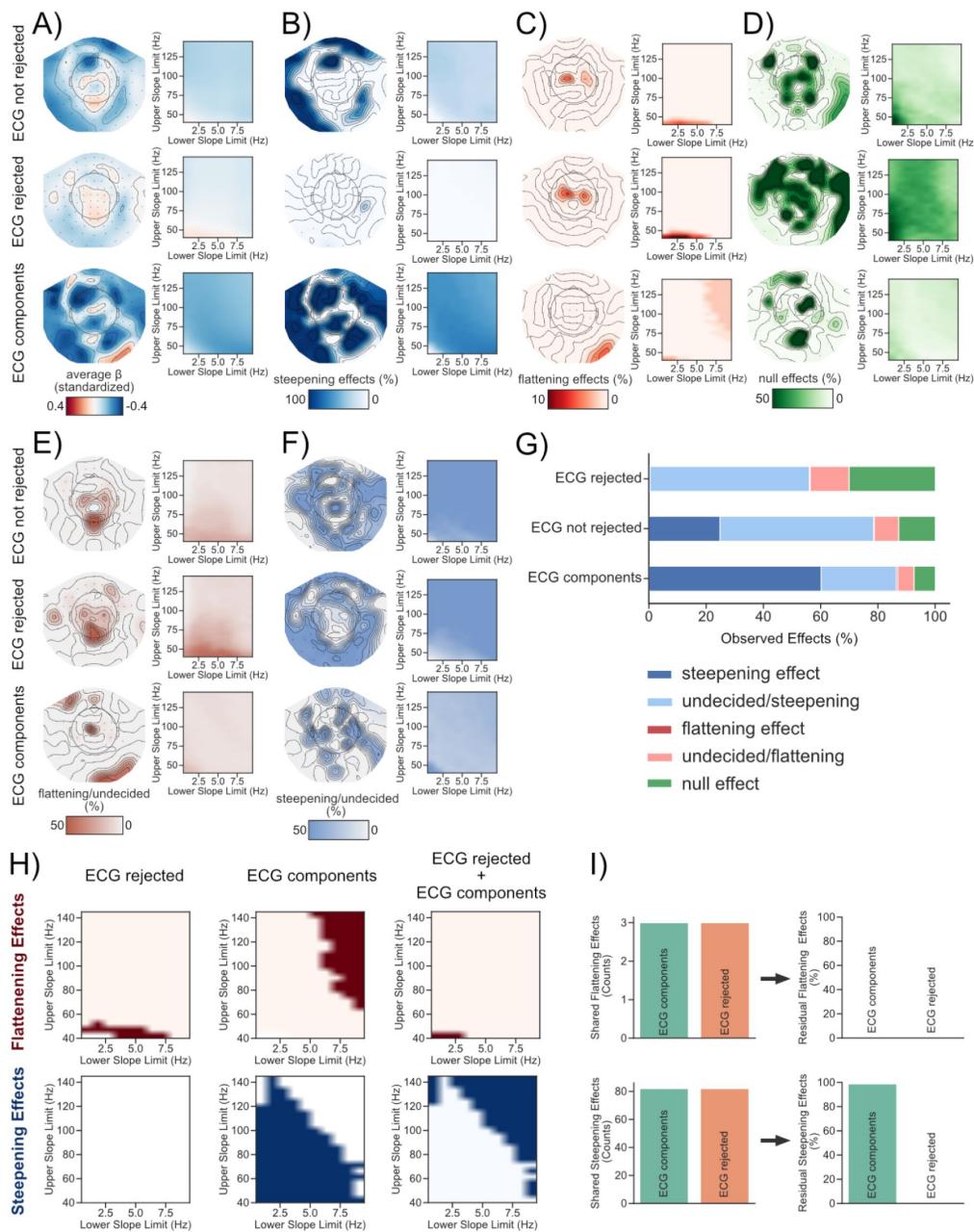


Figure 4

Age-related changes in aperiodic brain activity are most pronounced in cardiac components.

Age was used to predict the spectral slope at rest in three different conditions (ECG components not rejected, ECG components rejected and ECG components only) per channel across a variety of frequency ranges. **A)** Standardized beta coefficients either per channel averaged across all frequency ranges (left) or per frequency range (right) averaged across all channels. Age-related **B)** steepening, **C)** flattening and **D)** null effects in the spectral slope were observed and visualized in a similar manner as in **A)**. **E)** and **F)** We further show the direction of results where we didn't find enough evidence to support either a steepening, flattening or null effect. **G)** Summary of all observed findings in %. **H)** at some frequency ranges neural and cardiac aperiodic activity change independently with age (see also **BC)**. **I)** At other frequency ranges cardiac and neural aperiodic activity are similarly modulated by age. We used cardiac and neural aperiodic activity as predictors for age in a multiple regression model to test whether both explain unique variance in aging. This analysis reveals that when adding both $MEG_{ECG \text{ components}}$ and $MEG_{ECG \text{ rejected}}$ as predictors age-related flattening effects were reduced, yielding no longer significant flattening results (see **Figure 5B**; upper panel). However, in case of the observed steepening effects, significant effects for $MEG_{ECG \text{ components}}$ remained in 98.75% of the tested frequency ranges (see **Figure 5B**; lower panel).

not been reported in EEG recordings, which suggests that they may be in part linked to physiologically measured $1/f$ noise differently affecting magnetic and electric recording devices⁴³ (see Discussion).

However, despite the large amount of age-related steepening effects, we also noted age-related flattening in spectral slopes that occurred mainly at centrally and parietally located electrodes in lower frequency ranges between 0.5 and 45Hz (see **Figure 4C**). Importantly, these results overlap both in frequency range and recording site with some of the results previously reported in the literature^{8,44}. A majority of results fall in the category “undecided” (see **Figure 4EFG**) as there was not enough evidence to either support a steepening, flattening or null effect⁴⁰. Albeit undecided we still visualized the respective direction of these results labeling them either as “undecided/steepening” or “undecided/flattening” to give a descriptive overview of the associated spatial locations and frequency ranges in which these results were observed (see **Figure 4EF**). In sum, this analysis suggests that a flattening of spectral slopes with age can be observed at some of the previously reported frequency ranges (~ 0.5 –45Hz) and spatial locations (on central, parietal and occipital sensors). However, these results represented only 0.3% of effects across all processing settings (conditions, sensors and frequency ranges). Even when restrictively looking only at the investigated frequency ranges between 0.5 and 50Hz, only 1.2% (0.4% after maxfilter see **Supplementary S8**) of effects across these residual settings were showing an age related flattening of the spectral slope. This suggests that age-related flattening of the spectral slope is tied to specific recording sites and frequency ranges.

Age-related changes in aperiodic brain activity are linked to cardiac activity in a frequency dependent manner

So far we have shown that age-related steepening/flattening of the spectral slope in the MEG is both dependent on the investigated frequency range and the sensor selection. While a vast majority of our results indicate an age-related steepening of the spectral slope (in contrast to previous findings), we also noted a flattening of the spectral slope at a subset of central sensors in the lower frequency range (~ 0.5 –45Hz; in line with previous findings; ^{8,44}). Some of the observed age-related flattening and steepening effects were solely present in one of the tested conditions (see **Figure 4H**). This suggests that aperiodic brain activity ($\text{MEG}_{\text{ECG rejected}}$), at some frequency ranges, changes with age independently of cardiac activity ($\text{MEG}_{\text{ECG component}}$) and vice versa. However, we also noted shared effects at other frequency ranges (i.e. effects present both in the $\text{MEG}_{\text{ECG rejected}}$ and $\text{MEG}_{\text{ECG component}}$ condition; see **Figure 4I**).

To see if $\text{MEG}_{\text{ECG rejected}}$ and $\text{MEG}_{\text{ECG component}}$ explain unique variance in aging at frequency ranges where we noticed shared effects, we averaged the spectral slope across significant channels and calculated a multiple regression model with $\text{MEG}_{\text{ECG component}}$ and $\text{MEG}_{\text{ECG rejected}}$ as predictors for age (to statistically control for the effect of $\text{MEG}_{\text{ECG components}}$ and $\text{MEG}_{\text{ECG rejected}}$ on age). This analysis was performed to understand whether the observed shared age-related effects ($\text{MEG}_{\text{ECG rejected}}$ and $\text{MEG}_{\text{ECG component}}$) are (in-)dependent. The analysis revealed that when adding both $\text{MEG}_{\text{ECG component}}$ and $\text{MEG}_{\text{ECG rejected}}$ as predictors, age-related flattening effects were reduced, yielding no longer significant flattening results (see **Figure 4B**; upper panel). However, in case of the observed steepening effects, significant effects for $\text{MEG}_{\text{ECG components}}$ remained in 98.75% of the tested frequency ranges (see **Figure 4B**; lower panel). In sum, these results suggest that whether or not aperiodic brain activity changes independently from cardiac activity with age depends on the recording site and the selected frequency range.

Outlook

So far, we have shown that slow physiological changes (e.g. aging) can modulate aperiodic cardiac activity. To further understand the extent by which aperiodic ECG signals are also co-modulated by rapid event-related changes e.g. in cognitive tasks, we investigated the ECG recordings of a dataset

employing a working memory paradigm^{45,46} (see **Figure 5A**; for details *Methods - Working Memory Analysis*). Similarly, as in Donoghue et al.⁵ we compared a prestimulus “Baseline” to a post stimulus “Delay” period during a working memory task. Interestingly, akin to the EEG results reported by Donoghue et al.⁵, we observed a consistent flattening of the aperiodic slope for cardiac activity in the delay period (see **Figure 5BD**; $\beta_{\text{standardized}} = 0.23$, $HDI = [0.16, 0.32]$). Furthermore, upon comparing the change of slope relative to the baseline period across different levels of cognitive load we noticed that the flattening effect of the slope was modulated by cognitive load (see **Figure 5E**). The slope flattened the most in the condition with the highest working memory load (13 items; $\beta_{\text{standardized}} = 0.42$, $HDI = [0.33, 0.52]$), followed by the high load (9 items; $\beta_{\text{standardized}} = 0.39$, $HDI = [0.29, 0.48]$) and the low load condition (5 items; $\beta_{\text{standardized}} = 0.25$, $HDI = [0.15, 0.35]$). These results highlight the importance of considering the influence of cardiac activity when investigating changes in aperiodic activity in a state dependent manner.

Discussion

Aperiodic processes are ubiquitous in nature⁴⁷. They can be observed not only in physiological recordings but are also found in earthquakes, economics, ecology, epidemics, speech, and music^{47,48}. In measurements containing multiple aperiodic signals, aperiodic signals might even influence each other (e.g. neural speech tracking^{49,50}). The signals measured using M/EEG reflect a mixture of physiological sources (eg. cortical, cardiac, myographic and ocular), each of which exhibits aperiodic and periodic properties. To understand the (a-)periodic signal measured using M/EEG it is inevitable to understand how these different sources contribute to the (a-)periodic M/EEG signal. This becomes especially important when multiple physiological sources are modulated by the same traits, states and disorders. Cardiac and cortical recordings of aperiodic electrophysiological signals are related to age^{8,13}, cognitive states (eg. awake vs. sleep^{4,14}) and disorders such as Parkinson’s disease and epilepsy^{9,10,15,16}. So far cardiac and cortical activity were mostly analyzed separately (see **Figure 1C**). In the present study we investigated whether age-related changes in neural and cardiac aperiodic activity are (in-)dependent. Our results demonstrate that while cardiac activity significantly contributes to age-related changes in aperiodic activity, the extent of this influence varies by frequency range and sensor location—with neural aperiodic activity persisting in some instances. These findings, robust across diverse processing choices and replicable in an independent MEG dataset (see **Supplementary S1**), may also extend beyond aging. Notably, in a working memory paradigm, the aperiodic spectrum of “pure” ECG recordings is modulated by working memory load, mirroring previous findings⁵ and underscoring the potentially broader impact of cardiac activity on the aperiodic signal recorded using M/EEG.

Differences in aperiodic activity between magnetic and electric field recordings

Surprisingly, a vast amount of our results using MEG data indicate a steepening of the spectral slope with age. This is contrary to previous findings using mainly EEG/ECOG data⁸ that commonly show a widespread flattening of the spectral slope with age⁸. Similarly, we also noticed an age-related flattening on simultaneous ECG recordings (see **Figure 2C**). So do these discrepancies reflect general differences between electric vs. magnetic recordings of physiological activity? Previous research has shown scaling differences in the spectral slope between MEG and EEG recordings⁴³. These scaling differences are partly widespread (overall flatter sloped spectra in MEG data) and partly regionally-specific (steeper sloped spectra at vertex in MEG compared to EEG recordings and vice versa at frontal regions). These observations have been linked to non-resistive properties of tissue (i.e. the propagation of the electric field through tissue is frequency dependent^{43,51}). This differently affects the signal recorded using MEG and EEG, as magnetic field recordings are not distorted by the tissue conductivity of the scalp, skull, cerebrospinal fluid and brain⁵². An alternative, but not exclusive, hypothesis suggests that even under the

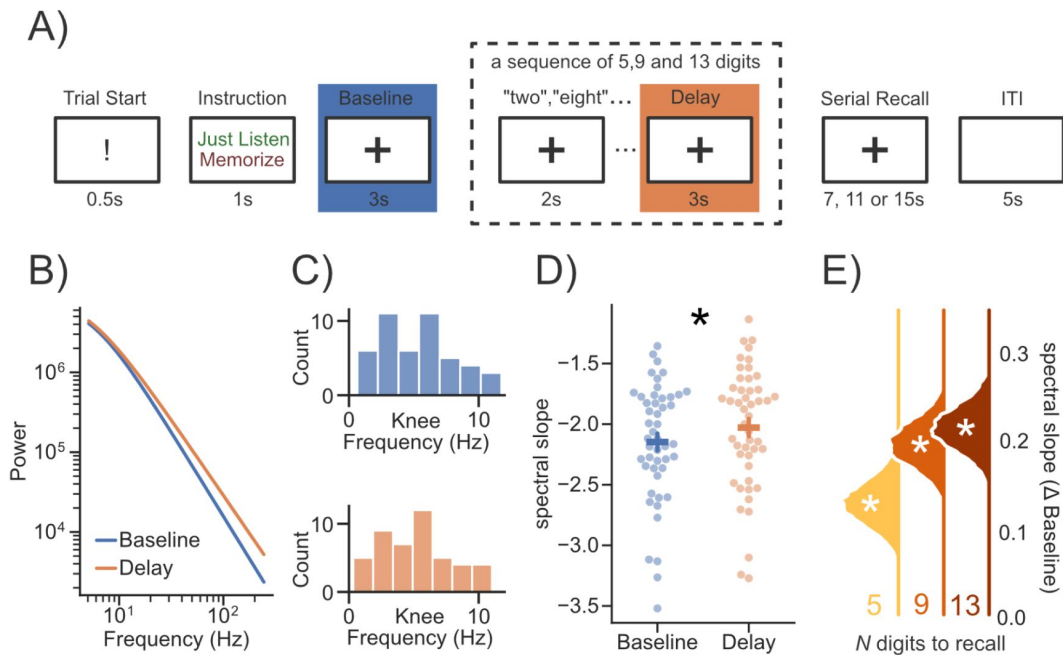


Figure 5

Event-related spectral parametrization of working-memory in the ECG.

A) Subjects were asked to either “listen” to or “memorize” a sequence of 5, 9 and 13 digits (adapted from [45](#), [46](#)). Spectra in the “Baseline” period were compared to the “Delay” period of the “memorize” condition. **B)** The averaged evoked difference in the aperiodic spectrum between baseline and delay periods. The spectra were reconstructed from the aperiodic parameters of the spectral fits and plotted as a function of frequency past the average knee frequency (~ 5 Hz; **C**). The spectral slope of the ECG signal was significantly flatter during the “Delay” compared to the “Baseline” period (**D**). **E)** The flattening of the spectral slope relative to “Baseline” was strongest in conditions with higher working memory load. Error bars indicate standard errors of the mean. A * indicates a “significant” effect (see Methods - Statistical Inference).

assumption of a purely resistive medium (which is unlikely^{53,54}), frequency scaling differences between MEG and EEG may emerge in relation to the space/frequency structure of the recorded activity⁵⁵. Under this hypothesis lower frequencies are suggested to involve synchronous activity in larger patches of cortex whereas higher frequencies involve synchronous activity in smaller cortical patches. Bénar et al.⁵⁵ demonstrate that EEG typically integrates activity over larger volumes than MEG, resulting in differently shaped spectra across both recording methods.

During aging both changes in conductive tissue properties^{56,57} and functional connectivity occur⁵⁸. Hypothetically, an interaction between several factors that differently affect MEG and EEG (e.g. age-related changes in non-resistive properties of tissue and in functional connectivity) may therefore potentially explain differently shaped spectra in MEG compared to EEG recordings. Future research is needed to explore the differences in magnetic and electric field recordings to understand the age-related changes to non-resistive tissue properties alongside age-related changes in functional connectivity. Differences in electric and magnetic field recordings aside, aperiodic activity may not change strictly linearly as we are ageing and studies looking at younger age groups (e.g. <22; ⁴⁴) may capture different aspects of aging (e.g. brain maturation), than those looking at older subjects (>18 years; our sample). A recent report even shows some first evidence of an interesting putatively non-linear relationship with age in the sensorimotor cortex for resting recordings⁵⁹. Another possible and not mutually exclusive explanation for the age-related steepening could be related to the ECG signal itself. We noticed an age-related steepening in the spectral slope of the ECG recording in the Cam-Can dataset between ~0.25 and 12Hz (see **Figure 2C**). Depending on how the power of the aperiodic ECG signal in this low frequency band is reflected on the MEG sensors, this could also bias the spectral slope of the combined MEG/ECG signal. However, an age-related steepening of the ECG was only noted at a frequency range between ~0.25 and 12 Hz making it an unlikely explanation of all the effects we detected in the MEG that span frequency ranges vastly exceeding the 0.25 - 12 Hz range.

Influences of preprocessing decisions on age-related changes in aperiodic activity

While differences between magnetic and electric field recordings may explain some observed differences in the widespread effects between electric and magnetic recordings of electrophysiological activity, we still observed flatter sloped spectra with age at a few MEG sensors across several frequency ranges (see **Figure 4CHI**). These findings are also in-line with previous analyses of MEG data investigating age-related changes in aperiodic activity^{60,61}. The frequency dependence of the flattening/steepening effects (see **Figure 4**) suggests that the slope of the power spectrum can be very sensitive to different preprocessing decisions that may emphasize different aspects of (neuro-)physiological activity. In case of the MEG signal this may include the application of Signal-Space-Separation algorithms (SSS^{24,62}), different thresholds for ICA component detection (see **Figure S7**), high and low pass filtering, choices during spectral density estimation (window length/type etc.), different parametrization algorithms (e.g. IRASA vs FOOOF) and selection of frequency ranges for the aperiodic slope estimation. We therefore applied a wide variety of processing settings when analyzing our data. Our results indicate overall steeper sloped spectra with increasing age across datasets and processing options for MEG. These observed steepening effects can be explained by cardiac activity (see **Figure 4HI**). Cardiac activity, in the form of the ECG, is also captured in EEG recordings (via volume conduction; see **Figure 3ABCD**). Our data suggests that the ECG signal is captured equally strong in concurrent MEG and EEG recordings (see **Figure 3ABCD**). Furthermore, separating ECG related components from neural activity using ICA seems to work worse in EEG compared to MEG recordings (see **Figure 3AB**). Difficulties in removing ECG related components from EEG signals via ICA might be attributable to various reasons such as the number of available sensors or assumptions related to the non-gaussianity of the underlying sources. Further understanding of this matter is highly important given that ICA is the most widely used procedure to separate neural from peripheral

physiological sources (see **Figure 1EF**). Additionally, it is worth noting that the effectiveness of an ICA crucially depends on the quality of the extracted components^{63,64} and even widely suggested settings e.g. high-pass filtering at 1Hz before fitting an ICA may not be universally applicable (see supplementary material of ⁶⁴).

Previous work^{12,65} has shown that a linked mastoid reference alone was particularly effective in reducing the impact of ECG related activity on aperiodic activity measured using EEG. However, it should be considered that depending on the montage, referencing can induce ambiguities to the measured EEG signal. Linked mastoid referencing for instance can distort temporal activity⁶⁶, which is unproblematic in studies focussing on activity on central electrodes^{12,65}, but should be considered when focusing on activity from other recording sites⁶⁷.

To better delineate cardiac and neural contributions when investigating aperiodic activity, ECG recordings should become more standard practice in EEG research. Additionally, further method development is needed to better separate cardiac from neural activity in M/EEG recordings.

(Neuro-)physiological origins of aperiodic activity

Aperiodic activity is present in recordings of different physiological signals, including neural⁴, cardiac¹¹, and muscle activity¹². Our study investigated age-related changes in aperiodic activity using MEG, and found that these changes vary depending on the frequency range and recording site. Specifically, some changes were found to be uniquely linked to either cardiac or brain activity, while others were present in both signals (**Figure 4H**). Interestingly, some of these shared effects could be attributed to cardiac activity (see **Figure 4I**). However, some effects appeared to be equally influential in explaining age-related changes in both cardiac and brain activity, such that the individual effects disappeared when analyzed jointly (see **Figure 4I**, upper panel).

These findings underscore the complexity of analyzing aperiodic activity, indicating that the aperiodic signal recorded non-invasively originates from multiple physiological sources. These shared effects are particularly interesting, as they suggest that a common mechanism across physiological signals exists that underlies age-related changes of aperiodic activity. In fact, neural and vascular processes are known to interact with each other⁶⁸. Cardiovascular activity, in the form of the ECG, is also captured in M/EEG recordings (via volume conduction; see **Figure 3ABCD**). How longitudinal changes in neural and cardiac processes influence age-related changes in aperiodic activity is an exciting research question. This could be investigated in future studies utilizing longitudinal recordings of joint cardiac and neural activity. Understanding the relationship between neural and cardiac aperiodic activity is essential not only for identifying common underlying processes, but also for improving our understanding of the individual generative mechanisms of cardiac and neural aperiodic activity.

For example, a current popular hypothesis states that the generative process underlying aperiodic neural activity is mainly attributed to differences in the ratio between excitatory (AMPA) and inhibitory (GABA) currents that influence the slope of the neural power spectrum⁴². Excitatory currents such as AMPA decay faster, than inhibitory currents like GABA. This means that flatter power spectra may be indicative for the presence of more excitatory than inhibitory currents and vice versa (steeper sloped power spectra⁴²). This theory is (in part) based on research showing that GABAergic drugs like propofol⁴² and Glutamatergic drugs like ketamine⁶⁹ modulate the slope of electrophysiologically measured power spectra. However, propofol and ketamine not only influence neural activity, but also influence heart rate variability (a core component of the ECG^{70,71}). So, are drug induced effects on the slope of the power spectrum (measured using surface electrodes) conflated by changes in cardiac activity? Previous work has shown that propofol induced changes to the spectral slope were still present in EEG recordings after using ICA to reject ECG components from the data⁶⁵. However, our results suggest that cardiac activity

remains in EEG signals even after separating cardiac from neural sources using an ICA with a very sensitive rejection criterion (see **Figure 3ABCD**). It is therefore plausible that drug induced effects on aperiodic “neural” activity can still be conflated by cardiac activity. Future work is needed to see to what extent drug induced changes in aperiodic neural activity can also be attributed to ECG signals. Similar caveats adhere to other findings of functional modulations in aperiodic signals in cognitive states (e.g. awake vs. sleep^{4,14}) and disorders like Parkinson’s disease and epilepsy^{9,10,15,16}. This calls for the initiation of - ideally - multicenter coordinated activities aimed to replicate 1/f aperiodic neural activity effects (e.g., induced by anesthetic drugs) while considering cardiac activity. Another pending research question lies in understanding whether our findings on non-invasive data also translate to data from invasive recordings. Given that cardiac activity is also captured on e.g. ECoG¹⁹, an influence is not unlikely depending on the strength of cardiac activity relative to the measured neural activity.

It is worth noting that, apart from cardiac activity, muscle activity can also be captured in (non-)invasive recordings and may drastically influence measures of the spectral slope⁷². To ensure that persistent muscle activity does not bias our results we used changes in head movement velocity as a control analysis (see **Supplementary Figure S9**). However, it should be noted that this is only a proxy for the presence of persistent muscle activity. Ideally, studies investigating aperiodic activity should also be complemented by measurements of EMG. Whenever such measurements are not available creative approaches that use the steepness of the spectral slope (or the lack thereof) as an indicator to detect whether or not e.g. an independent component is driven by muscle activity are promising^{72,73}. However, these approaches may require further validation to determine how well myographic aperiodic thresholds are transferable across the wide variety of different M/EEG devices.

While the present analysis focuses on aperiodic activity, our results might also translate to older findings focusing on “presumably” periodic neural activity in canonical frequency bands (e.g. delta, theta, alpha). Until recently aperiodic activity was often discarded as noise. Recently developed algorithms have opened up possibilities to separately analyze aperiodic and periodic activity^{5,74}. This has for instance revealed that previously suspected periodic age-related changes in alpha power may actually be attributable to differences in aperiodic activity^{5,27}. As we have shown that age-related changes in aperiodic activity are linked to cardiac activity it is possible that our results also translate to previous studies conflating periodic and aperiodic activity. However, whether or not periodic activity (after detection) should be detrended using approaches like FOOOF or IRASA still remains disputed, as incorrectly detrending the data may cause larger errors than not detrending at all⁷⁵.

Recommendations

Changes in aperiodic activity of peripheral and neural signals are co-modulated by similar traits, states and disorders. To better disentangle physiological and neural sources of aperiodic activity, we propose the following steps to (1) measure and (2) account for physiological influences.

1. Measure potential confounding physiological signals explicitly (e.g. ECG) and test whether there is an association between the respective (a-)periodic signal and the feature of interest (e.g. age). In case that the feature of interest co-modulates neural and physiological aperiodic activity it is necessary to account for this.

2. Reduce the influence of physiological signals on neural activity as much as possible. Currently, ICA can be used to at least reduce the impact of cardiac activity (see also **Figure 3**). However, separating physiological from neural sources using an ICA is no guarantee that peripheral physiological activity is fully removed from the cortical signal. Even more sophisticated ICA based methods that e.g. apply wavelet transforms on the ICA components may still not provide a good separation of peripheral physiological and neural activity.^{76,77} This turns the process of deciding whether or not an ICA component is e.g. either reflective of cardiac or neural activity into a challenging problem. For instance, when we only extract cardiac components using relatively high detection thresholds (e.g. $r > 0.8$), we might end up misclassifying residual cardiac activity as neural. In turn, we can't always be sure that using lower thresholds won't result in misinterpreting parts of the neural effects as cardiac. Both ways of analyzing the data can potentially result in misconceptions. In the present study, we show that our effects are largely consistent across different thresholds (see **Supplementary Figure S7**), but future research should be devoted to developing objective criteria that can be used to make informed decisions, when results are inconsistent. Additionally, it might be necessary to invest in the development of new methods to better separate peripheral from neural signals, for instance combinations of ICA with other methods such as e.g. empirical mode decomposition (see^{78,79}). Other promising approaches may potentially involve bipolar referencing for EEG or spatial referencing approaches such as current source density.⁸⁰ How these approaches impact aperiodic activity should be further investigated. For MEG source reconstruction approaches like beamforming may be promising (see e.g. for reduction of tACS artifacts in the MEG⁸¹). In the case that it is not possible to sufficiently reduce the influence of physiological signals on neural activity it is necessary to control for this in the statistical model (e.g. by using the confounding physiological aperiodic signal as a covariate).

Conclusion

The present study, focusing on age-related changes in aperiodic neural and cardiac activity indicates that the aperiodic signal recorded using surface sensors/electrodes originates from multiple physiological sources. Cardiac and neural age-related changes in aperiodic activity vary depending on the frequency range and recording site. Conflating cardiac and neural contributions to aperiodic activity obstructs our understanding of both neural and cardiac aperiodic processes and should be avoided. These results highlight the need for concurrent recordings of cardiac and neural activity to further increase our understanding of both cortical and cardiac aperiodic activity and its association with age, cognitive states, and disorders.

Methods

Sample

The present study builds on resting-state data from several sources. ECG Datasets 1 & 2 were obtained from PhysioNet⁸² containing 1121 healthy volunteers between 18 and 92 years of age (divided into 15 age groups for anonymization purposes)⁸³. The MEG data analyzed in this study was obtained from two different sources: the Cam-CAN repository (main manuscript^{28,29}) and resting-state data routinely recorded at the University of Salzburg (**Supplementary Figure S5**). The sample obtained from Cam-CAN contained 655 healthy volunteers with an age range between 18 and 88 years of age with an average age of 54 and a standard deviation of 18 years. The sample obtained from the University of Salzburg contained 684 healthy volunteers with an age range between 18 and 73 years of age with an average age of 32 and a standard deviation of 14 years. ECG was recorded alongside all MEG recordings. Notably the age distribution recorded at the University of Salzburg was bimodal (with 423 subjects being below 30, the sample does not reflect an age-diverse population; see **Supplementary Figure S5**). For the ECG data, no specific

exclusion criteria for participants was applied. Data from MEG/EEG participants were excluded when no independent heart component ($N_{Cam-Can} = 18$) equal or above the threshold was found ($r > 0.4$; see MEG/EEG Processing - pre-processing). Furthermore, when the automated processing procedure resulted in errors, data was considered as missing. This resulted in a total of 1104 ECG recordings from PhysioNet, 627 MEG recordings from Cam-CAN and 655 MEG recordings from the University of Salzburg.

Literature analysis

The literature analysis was performed using the Literature Scanner (LISC)²¹ toolbox and custom written python functions. Briefly, LISC allows for the collection and analysis of abstracts and meta information from scientific articles through a list of search terms. In this manuscript, we used lists of terms for aperiodic activity (e.g. 1/f, power law, scale-free, etc.), recording devices (ECG, MEG and EEG) and related association terms (e.g. aging, working memory etc.) with relevant synonyms. The articles ($N_{Articles} = 489$) that reference these terms in their abstracts were extracted from the PubMed database. We extracted the proportion of articles related to aperiodic activity and associated with ECG and/or M/EEG (see **Figure 1A**). Furthermore, we displayed the publishing dates of articles in ECG or M/EEG over time (see **Figure 1B**). Additionally, we extracted the article counts of several association terms (e.g. aging) and visualized them separately for ECG and M/EEG. We then extracted the doi's of the articles associated with MEG and EEG separately and sequentially extracted the HTML of the full-texts (whenever the texts were accessible). This resulted in 213 articles for EEG and 66 articles for MEG. Afterwards we extracted search words related to ECG and cardiac activity (with relevant synonyms and word stems; cardio, cardiac, heart, ecg) from each manuscript along the respective word context (400 signs before and 600 after each search term; for each time a search word was mentioned in one of the extracted manuscripts). Manuscripts in which the word stems “reject” (and related synonyms and word stems; remov, discard, reject) were mentioned in one of the word contexts were temporally marked as “valid”. The word contexts were further queried for search terms related to common blind source separation artifact rejection approaches such as independent component analysis (ICA²²), singular value decomposition (SVD²³), signal space separation (SSS²⁴), signal space projections (SSP²⁵) and denoising source separation (DSS²⁶). All valid word contexts were then manually inspected by scanning the respective word context to ensure that the removal of “artifacts” was related specifically to cardiac and not e.g. ocular activity or the rejection of artifacts in general (without specifying which “artifactual” source was rejected in which case the manuscript was marked as invalid). This means that the results of our literature analysis likely present a lower bound for the rejection of cardiac activity in the M/EEG literature investigating aperiodic activity. Finally, we visualized the proportion of articles in relation to the respective search words (see **Figure 1DE**). Furthermore, we arbitrarily selected 60 articles investigating aperiodic activity and visualized the investigated frequency ranges alongside their respective upper and lower bounds (1FGH).

Statistical Inference

To investigate the relationship between age and aperiodic activity recorded using MEG, we used bayesian generalized linear models (GLMs) either built directly in PyMC (a python package for probabilistic programming⁸⁴) or in Bambi (a high-level interface to PyMC⁸⁵). Decisions for either Bambi or PyMC were made based on the accessibility of appropriate statistical families for the respective dependent variables in a GLM. Priors were chosen to be weakly informative⁸⁶ (exact prior specifications for each model can be obtained from the code in the corresponding authors github repository; see Data and Code availability).

Results were considered statistically significant if 94% of the highest (probability) density interval (HDI) of the posterior for a given standardized β -coefficient or (partial) correlation coefficient was not overlapping with a region of practical equivalence between -0.1 and 0.1 as suggested by Kruschke⁴⁰ based on negligible effect sizes according to Cohen (1988). Furthermore, it was

ensured that for all models there were no divergent transitions ($R_{\text{hat}} < 1.05$ for all relevant parameters) and an effective sample size > 400 (an exhaustive summary of bayesian model diagnostics can be found in [88](#)).

MEG/EEG Processing

MEG/EEG Processing - Data acquisition

MEG data (Cam-CAN; **Figure 2ABCD** [3](#), 4 & 5) was recorded at the University of Cambridge, using a 306 VectorView system (Elekta Neuromag, Helsinki). MEG data (Salzburg; **Figure 3** [3](#) & **Supplementary Figure S5** [3](#)) was recorded at the University of Salzburg, using a 306 channel TRIUX system (MEGIN Oy, Helsinki). Both systems are equipped with 102 magnetometers and 204 planar gradiometers and positioned in magnetically shielded rooms. In order to facilitate offline artifact correction, electrooculogram (VEOG, HEOG) as well as ECG was measured continuously in both recording sites. In a subset of the Salzburg recordings EEG was measured additionally (see **Figure 3ABCD** [3](#)) using a 32-channel system provided by the MEG manufacturer. Data recorded at Cambridge was online filtered at 0.03-330 Hz, whereas data recorded at Salzburg was online filtered at 0.1-333 Hz with a 1000 Hz sampling rate at both recording sites. Five Head-Position Indicator coils were used to measure the position of the head. All data used in the present study contains passive resting-state measurements lasting about ~8min (Cambridge) and ~5min (Salzburg). Further data processing at both recording sites was conducted similarly and will therefore be reported together.

MEG/EEG Processing - pre-processing

Initially, a Signal-Space-Separation (SSS^{[24](#),[62](#)}) algorithm was used to find and repair bad channels (implemented in MNE-Python *version 1.2*^{[89](#)}). The data was further processed by either not applying further SSS cleaning (main manuscript) or by applying an SSS algorithm for additional data cleaning (**Supplementary Figure S6** [3](#) & **S8**; implemented in MNE-Python *version 1.2*^{[89](#)}). The data were afterwards high-pass filtered at 0.1Hz using a finite impulse response (FIR) filter (Hamming window). EEG data was re-referenced to the common average (common practice in the field^{[67](#)}). For extracting physiological “artifacts” from the data, 50 independent components were calculated using the *fastica* algorithm^{[22](#)} (implemented in MNE-Python *version 1.2*; with the parallel/symmetric setting; note: 50 components were selected for MEG for computational reasons for the analysis of EEG data no threshold was applied). As ICA is sensitive to low-frequency drifts, independent components were calculated on a copy of the data high-pass filtered at 1Hz. Components related to cardiac and ocular activity were determined via correlation with concurrent ECG, VEOG and HEOG recordings. A threshold of $r > 0.4$ was applied to detect the ECG components and a threshold $r > 0.8$ for detecting EOG components in the data. The more sensitive threshold used for ECG component detection was decided upon based on the strong presence of ECG signals in resting state M/EEG recordings (see **Figure 2EFGH** [3](#)). The computed ICA was then applied to the original data, either rejecting all components apart from those related to the ECG, or rejecting only EOG related components, or ECG and EOG related components. This resulted in three conditions MEG_{ECG not rejected}, MEG_{ECG rejected} and MEG_{ECG component}. The data were then split into 2 second epochs and residual artifacts were determined using an adaptive and automatic artifact detection method (the “Riemannian Potato” implemented in *pyriemann*^{[90](#)}). Epochs were rejected when the covariance matrix of an epoch differed by > 2.5 standard deviations from a centroid covariance matrix.

MEG/EEG Processing - Temporal Response Functions

To estimate the extent at which ECG activity is captured by MEG/EEG recordings, we calculated temporal response functions (TRFs). In brief, the assumption behind a TRF is that a dependent variable is the result of one (or several) predictor variables. This approach can be used to model a

time-series as a linear function of one (or several) time series and can be formulated (for a single predictor⁹¹) as:

$$y_t = \sum_{\tau=\tau_{min}}^{\tau_{max}} h_{\tau} x_{t-\tau}$$

Where h represents the TRF, sometimes described as filter kernel, and τ represents

potential delays between y and x (for an extensive explanation of the algorithm used herein see ⁹¹). Typically, this approach is used to estimate spectro-temporal receptive fields, e.g. in response to auditory stimuli⁹², where the interpretation of a TRF follows that of an ERP/F in response to a continuous input signal⁹³. This model can be used as a forward or encoding model to predict the brain response from a stimulus representation or as a backward or decoding model to reconstruct a stimulus representation from brain recordings. We used this approach here to detect whether ECG and M/EEG influence each other. Concurrent recordings of MEG and EEG data alongside ECG were only available for a subset of subjects ($N = 20$). We therefore selected those subjects from the subjects data pool of the Salzburg sample to perform the TRF analysis. Forward and Backward models were calculated between the M/EEG signal and the signal recorded using ECG. This was done to see if signals measured using M/EEG and ECG follow/precede each other (non zero-time lags; e.g. via interoception see⁴¹) or if the signals are instantaneously related (zero-time-lags; therefore likely corresponding to the same underlying signal source). As computations of TRFs are memory and time extensive, the MEG, EEG and ECG data were additionally low pass filtered at 45Hz (to avoid aliasing) and downsampled to 100Hz before the analysis (a common practice when computing TRFs; see ⁹¹). For the computation of the TRFs, the ECG and M/EEG data were normalized (z-scored), and an integration window from -250 to 250 ms with a kernel basis of 50 ms Hamming windows was defined. To prevent overfitting, model parameters were adjusted using a four-fold nested cross validation (two training folds, one validation fold, and one test fold), each partition served as a test set once. The accuracy of the model was assessed by calculating the Pearson correlation coefficient between the respective predicted and measured time series. We calculated the same model in the forward direction (encoding model; i.e. predicting M/EEG data in a multivariate model from the ECG signal) and backward direction (decoding model; i.e. predicting the ECG signal using all M/EEG channels as predictors). The accuracy of the decoding model was used in **Figure 3CD** to assess how well the ECG time series was decodable from M/EEG data. For the respective encoding model we visualized the non-normalized encoding weights (see **Figure 3AB**). The TRF time courses visualized in **Figure 3A** were obtained by computing a principle component analysis (PCA⁹⁴) within subject and across all channels, whereas the distribution of peaks visualized in **Figure 3B** was obtained by calculating the root mean square across channels and extracting the maximum value.

MEG/EEG Processing - Spectral analysis

Power spectra were computed using Welch's method⁹⁵ between 0.1 and 145Hz (0.5 Hz resolution). Aperiodic activity was extracted using the IRASA method⁷⁴ implemented in the YASA package⁹⁶. Furthermore, in addition to the main model fit between 0.1-145Hz, additional slopes were fitted to the aperiodic spectrum in 5Hz steps starting from 45Hz as upper frequency limit and in 1Hz steps from 0.5Hz to 10Hz as lower frequency limit. Additionally, to investigate the robustness of our result the spectral parameterization algorithm (implemented in *FOOOF* version 1.0.0⁵) was used to parametrize raw power spectra. Power spectra were parameterized across frequency ranges of 0.5–145 Hz. FOOOF models were fit using the following settings: peak width limits: [1 – 6]; max number of peaks: 2; minimum peak height: 0.0; peak threshold: 2.0; aperiodic mode: 'fixed'. Goodness of fit metrics for both IRASA and FOOOF can be found in **Supplementary S2**.

ECG Processing

The ECG data recorded as part of the MEG recordings were processed alongside the MEG data. Therefore the pre-processing and spectral analysis settings from the section “MEG/EEG Processing” also apply to the ECG aspect of datasets 3 & 4 (see [Figure 2](#)). Below, ECG processing for the data obtained from PhysioNet are described.

ECG Processing - Data acquisition

The ECG data obtained from PhysioNet were acquired at the Jena university hospital ⁸³. The study was approved by the ethics committee of the Medical Faculty of the Friedrich Schiller University Jena. All research was performed in accordance with relevant guidelines and regulations. The informed written consent was obtained from all subjects. ECG data were recorded at a sampling rate of 1000 Hz using one of two different recording devices. Either an MP150 (ECG100C, BIOPAC systems inc., Golata, CA, USA) or a Task Force Monitor system (CNSystems Medizintechnik GmbH, Graz AUT). More detailed information about the ECG recordings can be obtained from physionet.org. The data were further analyzed using spectral analysis.

ECG Processing - Spectral analysis

Power spectra were computed using Welch’s method⁹⁵ implemented in *neurodsp*⁹⁷ between 0.25 and 145Hz (0.1 Hz resolution). The spectral parameterization algorithm (implemented in *FOOOF* version 1.0.0⁵) and IRASA (implemented in the YASA package⁹⁶) were then used to parametrize the power spectra. Power spectra were parameterized across a frequency range of 0.25–145 Hz using the same settings specified in “MEG/EEG Processing – Spectral analysis”). However, the aperiodic mode was set to ‘knee’ given that the power spectra (on average) showed a clear ‘knee’ in log-log coordinates (see [Supplementary Figure S1](#)). Additionally, we fitted several slopes to the aperiodic spectrum split in a lower range (0.25 – 20 Hz) and a higher range (10 – 145 Hz). The split in low and high frequency range was performed to avoid spectral knees at ~15 Hz in the center of the slope fitting range.

ECG Processing - Heart rate variability analysis

Heart rate variability (HRV) was computed using the NeuroKit2 toolbox, a high level tool for the analysis of physiological signals. First, the raw electrocardiogram (ECG) data were preprocessed, by highpass filtering the signal at 0.5Hz using an infinite impulse response (IIR) butterworth filter(order=5) and by smoothing the signal with a moving average kernel with the width of one period of 50Hz to remove the powerline noise (default settings of `neurokit.ecg.ecg_clean`). Afterwards, QRS complexes were detected based on the steepness of the absolute gradient of the ECG signal. Subsequently, R-Peaks were detected as local maxima in the QRS complexes (default settings of `neurokit.ecg.ecg_peaks`; see ⁹⁸ for a validation of the algorithm). From the cleaned R-R intervals, 90 HRV indices were derived, encompassing time-domain, frequency-domain, and non-linear measures. Time-domain indices included standard metrics such as the mean and standard deviation of the normalized R-R intervals, the root mean square of successive differences, and other statistical descriptors of interbeat interval variability. Frequency-domain analyses were performed using power spectral density estimation, yielding for instance low frequency (0.04–0.15Hz) and high frequency (0.15–0.4Hz) power components. Additionally, non-linear dynamics were characterized through measures such as sample entropy, detrended fluctuation analysis and various Poincaré plot descriptors. All these measures were then related to the slopes of the low frequency (0.25 – 20 Hz) and high frequency (10 – 145 Hz) aperiodic spectrum of the raw ECG.

Working Memory Analysis Sample

As an outlook to which extent the present findings, focussed on aging, may translate to studies investigating changes in “cortical” aperiodic in other settings (e.g. in a state dependent) we analyzed data from a working memory paradigm^{45,46} (see [Figure 5A](#) for an overview of the experimental paradigm). The original study included 86 healthy volunteers, due to technical difficulties ECG and EEG recordings were only available for a subset of subjects. In the present study we analyzed the data of 48 Subjects. The age-range of the investigated sample was between 18 and 24 years with an average age of 20 years. Informed consent was obtained from each participant and the experimental protocol was approved by the Ural Federal University ethics committee.

Digit span task

Each trial began with an exclamation mark for 0.5 s along with a recorded voice command “begin” – indicating the start of the trial. The exclamation mark was followed by an instruction to either memorize the subsequent digits in the correct order (memory condition) or to just listen to the digits without attempting to memorize them (control condition). The instruction was followed by a three-second “Baseline” period. Then either 5, 9, or 13 digits were presented auditorily with an interstimulus interval of 2 seconds. The digits were presented with a female voice in Russian. Each of the digits from 0 to 9 was used, and the mean duration of each digit was 664 ms (min: 462 ms, max: 813 ms). The last digit in the sequence was followed by a 3-sec “Delay” period. During the baseline, encoding, and “Delay” period, participants were fixating a cross (1.2 cm in diameter) on the screen. In the memory condition, the participants were asked to recall each digit out loud in the correct order starting from the first one (i.e., serial recall). The retrieval was recorded by a computer microphone controlled by PsychoPy⁹⁹. The participants had 7, 11, and 15 seconds for 5, 9, and 13 digit sequences, respectively, to recall the digits. The retrieval was followed by an inter-trial interval of 5 s. In the control condition (passive listening), presentation of the digits and the “Delay” period was followed immediately by an inter-trial interval of the same duration. There were 9 blocks in total with 54 passive listening and 108 memory trials overall. Each block consisted of 3 control (one of each load) followed by 12 memory (4 trials on each level of load, in random order) followed again by 3 control trials. Before the main working memory task, each participant completed 6 practice trials (3 passive listening and 3 memory trials).

Data acquisition

ECG was recorded from one channel with the active electrode placed on the right wrist and the reference electrode on the left wrist, and the ground on the left inner forearm at 3 cm distally from the elbow.

Data analysis

The continuous ECG data was high-pass filtered at 0.1Hz using a finite impulse response (FIR) filter (Hamming window) and downsampled to 500Hz after applying an anti-aliasing filter. The downsampled data was then split in 3 second epochs separately for the “Baseline” and “Delay” periods, as well as the amount of presented digits (5, 9 or 13). The epoched data was further analyzed either by calculating power spectra over the data split in the “Baseline” and “Delay” condition irrespective of the amount of presented digits ([Figure 5BCD](#)) or by retaining the information about the amount of presented digits ([Figure 5E](#)). Power spectra were computed using Welch’s method⁹⁵ between 0.1 and 245 Hz (0.333 Hz resolution). The spectral parameterization algorithm (implemented in *FOOOF* version 1.0.0⁵) was then used to parametrize the power spectra. Power spectra were parameterized across a frequency range of

0.1–245 Hz using the same settings specified in “MEG/EEG Processing - Spectral analysis”). However, the aperiodic mode was set to ‘knee’ given that the power spectra (on average) showed a clear ‘knee’ in log-log coordinates.

Statistical analysis

To investigate the relationship between working memory load and aperiodic activity recorded via ECG we implemented bayesian linear mixed effect models in Bambi⁸⁵ using the following formulas according to the Wilkinson notation¹⁰⁰ with the “Baseline” condition set as *Intercept* in the following models.

spectral slope ~ 1 + Delay + (1 | subject id) [Figure 5D]

spectral slope ~ 1 + N Digits to recall + (1 | subject id) [Figure 5E]

Data Visualization

Individual plots were generated in python using matplotlib, seaborn, and mne-python. Plots were then arranged as cohesive figures with affinity designer (<https://affinity.serif.com/en-us/designer/>).

Data availability

The data analyzed in the main manuscript are mostly obtained from open data sources. The ECG data (Dataset 1 & 2; Figure 2) were obtained from physionet.org. The MEG/EEG dataset (Figure 3) was obtained at the University of Salzburg as part of routine resting state MEG recordings and is available upon request. The data for the MEG analysis in the main manuscript (Figure 4) are obtained from cam-can.org. The data for the working memory analysis was obtained from openneuro.org (Figure 5).

Code availability

All code used for the analysis is publicly available on GitHub at: https://github.com/schmidtfa/cardiacc_1f & https://github.com/schmidtfa/ecg_1f_memory.

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Additional information

Author contributions

Conceptualization: FS, NW Formal Analysis: FS, SKD Investigation: FS Methodology: FS, GD Software: FS

Supervision: NW, GD Visualization: FS Writing—original draft: FS

Writing—review & editing: FS, SKD, ET, NW, GD, DPK

Supplementary material

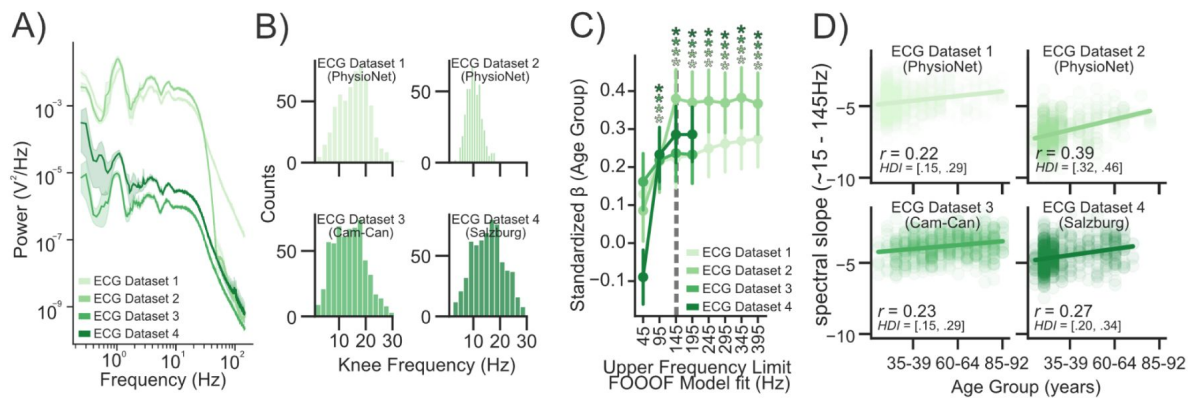
Supplementary Text S1 - Control Analyses: Age-related steepening of the spectral slope in the MEG

First, we conducted a median split of age to compare the raw power spectra averaged across channels (see **Supplementary Figure 2A** [↗](#)). This shows that on the grand average across channels the spectral slope was slightly steeper in older subjects even before spectral parametrization. Furthermore, the use of blind source separation artifact rejection approaches may influence power spectral densities by reducing external noise in the signal. We therefore investigated whether and how the use of a Signal-Space-Separation algorithm (SSS^{24,62} [↗](#)) and different ICA thresholds influence the reported results. On the grand average across sensors slightly stronger steepening effects were observed for the MEG_{ECG} not rejected data compared to MEG_{ECG} rejected when not cleaning the data using SSS and vice versa (see **Supplementary Figure S6** [↗](#) for a comparison). On the level of single sensors the application of SSS resulted in a further reduction of both flattening and steepening effects in all conditions except for MEG_{ECG} components, where we noted a 20% increase in steepening effects (see **Supplementary Figure S7** [↗](#)). Considering that we detected less and weaker aperiodic effects when using SSS maxfilter is it now advisable to omit maxfilter, when analyzing aperiodic signals? We don't think that we can make such a judgment based on our current results. This is because it's unclear whether or not the reduction of effects stems from an additional removal of peripheral information (e.g. muscle activity; that may be correlated with aging) or is induced by the SSS maxfiltering procedure itself. As the use of maxfilter in detecting changes of aperiodic activity was not subject of analysis that we are aware of, we suggest that this should be the topic of additional methodological research.

To ensure that the effects shown are not dependent on the ICA thresholds we used, an analysis predicting the grand average spectral slope based on age was also conducted for other correlation thresholds showing an overall similar pattern (see **Supplementary Figure S6** [↗](#)). We further managed to replicate the finding that age has the strongest impact on the spectral slope of the ECG components using a different algorithm to extract aperiodic activity (FOOOF⁵; see **Supplementary Figure S2** [↗](#) & S3). Using FOOOF⁵, we also investigated the impact of different slope fitting options (fixed vs. knee model fits) on the aperiodic age relationship (see **Supplementary Figure S4** [↗](#)). The results that we obtained from these analyses using FOOOF offer converging evidence with our main analysis using IRASA. We further replicated our findings on an additional dataset containing resting state recordings ($N = 655$) obtained as part of MEG studies routinely conducted at the University of Salzburg (see **Supplementary Figure S4** [↗](#)).

While all these control analyses indicate that the age-related steepening effects occur robustly in the MEG it is unclear whether they can be exclusively attributed to cardiac activity. Crucially, the topography of the observed steepening effects is present across the scalp and prominent at frontal and temporal sensors around the MEG helmet (albeit also observable at central locations; see MEG_{ECG} Components). This topography is suggestive of artifacts induced by muscle activity (e.g. head/eye movements). We therefore used the subject's head movement information obtained via continuous hpi measurements as a covariate (i.e. 5 coils continuously emitting sinusoidal waves at 293 Hz, 307 Hz, 314 Hz, 321 Hz and 328 Hz to localize the head position in the scanner). While head movements increased significantly with aging ($\beta_{\text{standardized}} = 0.23$, $HDI = [0.18, 0.28]$, see **Supplementary Figure S8** [↗](#)) it was not sufficient to explain the observed steepening or flattening effects in the spectral slope (see **Supplementary Figure S8** [↗](#)). We further investigated age-related changes to the spectral slope of the vertical and horizontal EOG channels indicating no significant age-related steepening/flattening across the investigated frequency ranges (see **Supplementary Figure S8**). Surprisingly, all these results indicate an age-related steepening in the spectral slope of MEG data both when averaged across sensors and on most individual sensors across two large

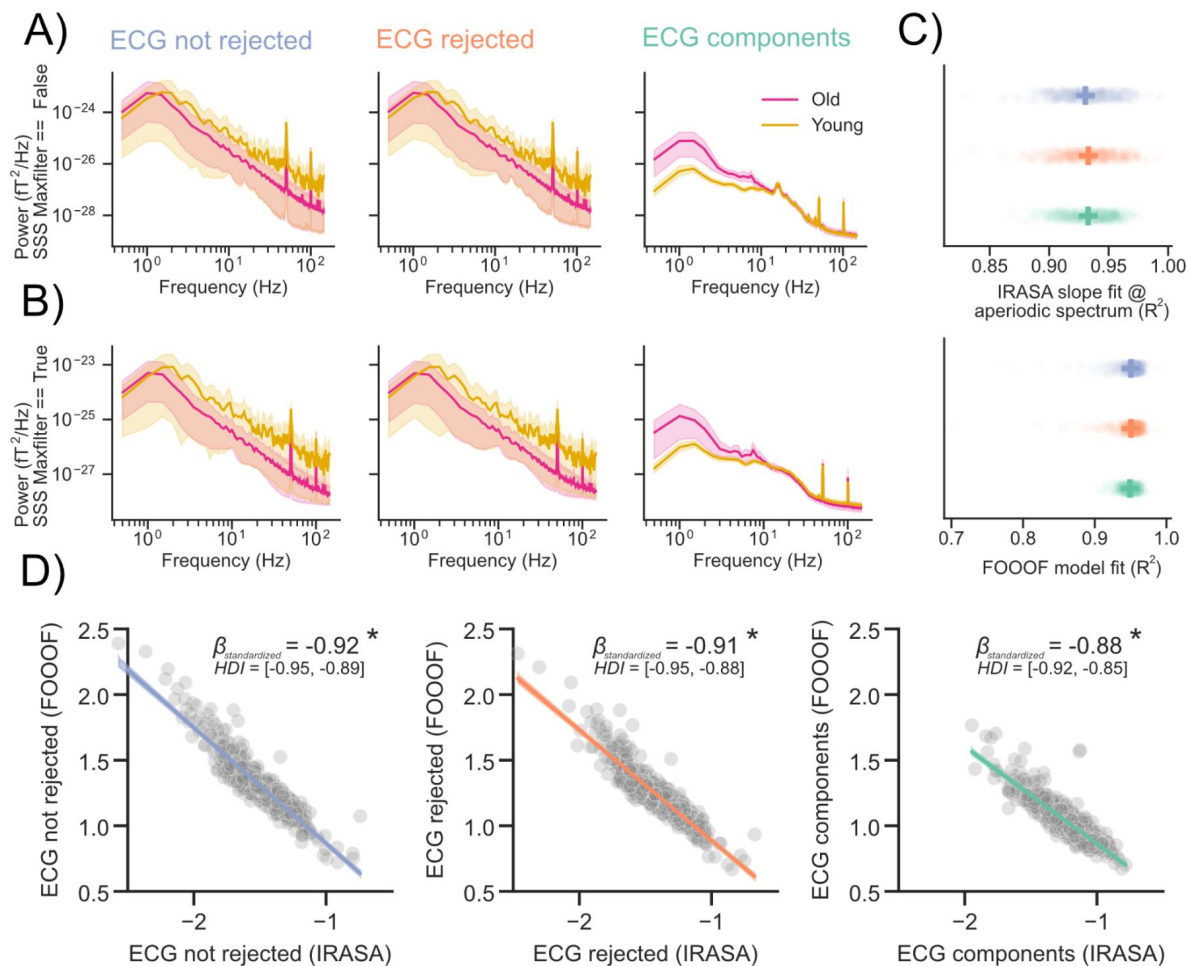
datasets. This finding is contrary to previous findings showing a flattening of spectral slopes with age in recordings of brain activity⁸ and cardiac activity¹³ (see also **Figure 2C**). This discrepancy can potentially be explained by multiple factors including, physiologically measured 1/f noise differently affecting magnetic and electric recording devices⁴³, preprocessing choices, fitting ranges for the 1/f slope, electrode selection etc. (see discussion).



Supplementary Figure S1.

ECG Spectra + Knee Frequency:

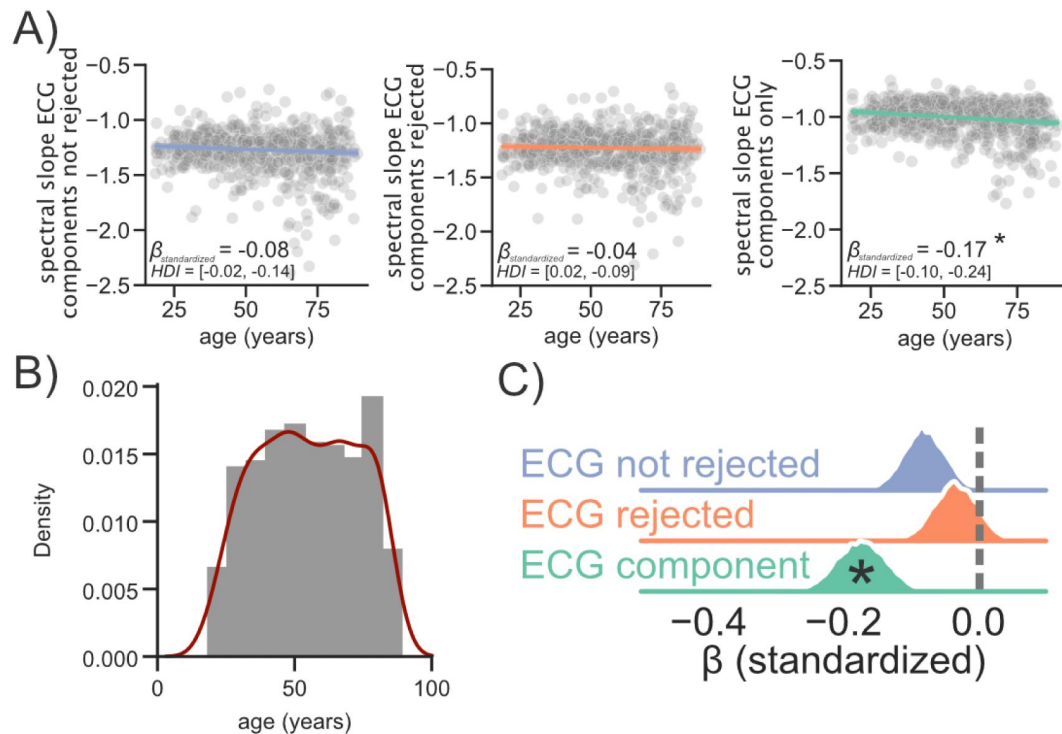
Aperiodic signals recorded using ECG can be associated with aging. **A)** grand average power spectra plotted separately per Dataset **B)** indicating a strong "knee" from ~15Hz. **CD)** The spectral slope recorded using ECG was calculated using different upper frequency limits and correlated with age. This analysis shows that the association between age and spectral slope increased until ~145 Hz in 3 of 4 datasets.



Supplementary Figure S2.

Raw Spectra + Goodness of fit metrics:

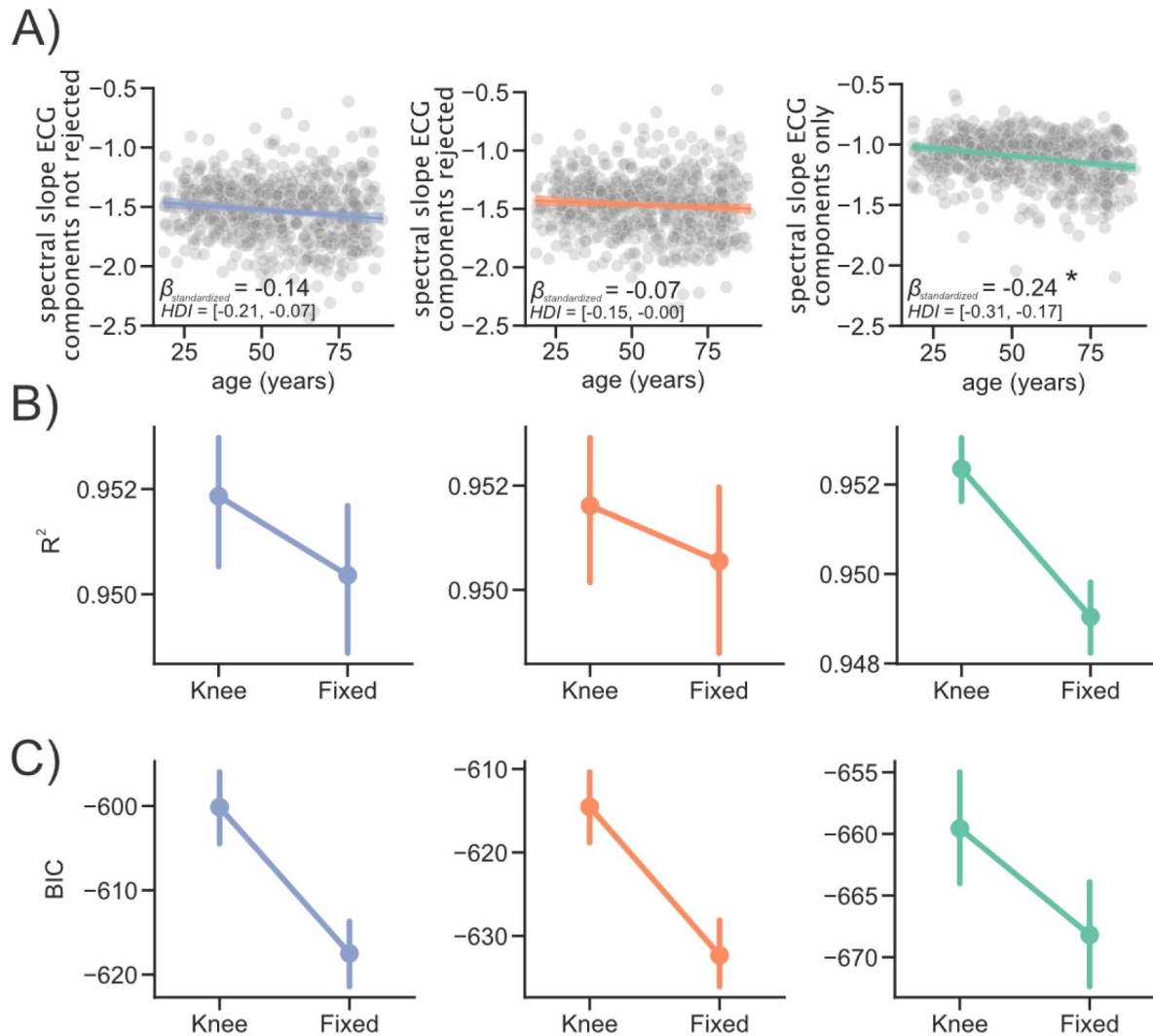
Grand average power spectra for the MEG data recorded at Cambridge and split in the three conditions MEG_{ECG not rejected}, MEG_{ECG rejected}, MEG_{ECG components}. **A)** Spectra averaged across channels split in older and younger subjects (median split). **B)** Aperiodic activity was extracted from the power spectra using the IRASA method [74](#) implemented in the YASA package [96](#). * indicate a “significant” effect (see Methods - Statistical Inference). **C)** Goodness of fit was assessed using R^2 for both the IRASA and FOOOF model fit (see [Supplementary S2](#) /S3 for FOOOF fits). **D)** The spectral exponent obtained from FOOOF was compared to the spectral slope extracted from IRASA showing that both are highly related.



Supplementary Figure S3.

Replication FOOF:

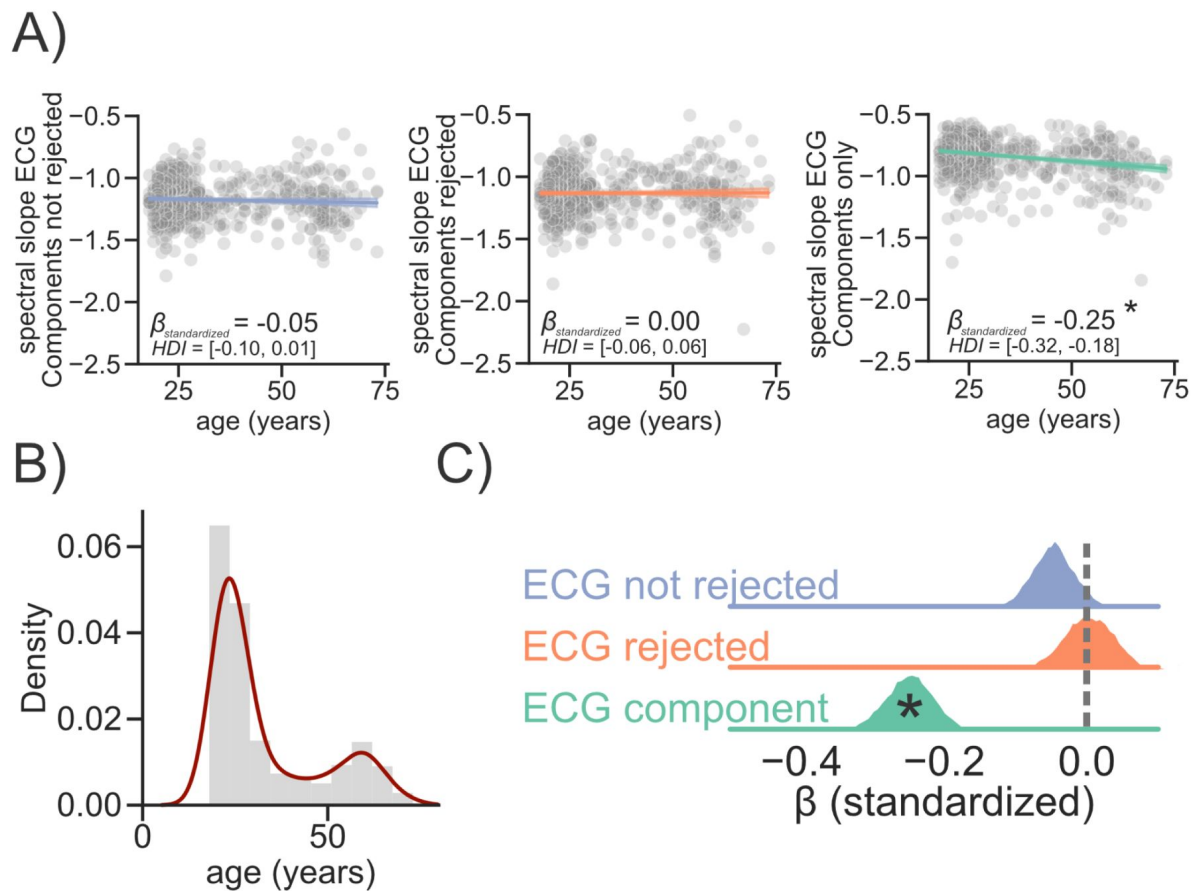
Age-related changes in aperiodic brain activity can be explained by cardiac components **A)** Age was used to predict the spectral slope at rest in three different conditions (ECG components not rejected [blue], ECG components rejected [orange], ECG components only [green]. **B)** Age distribution in the sample recorded at the University of Cambridge. **C)** Comparison of standardized beta coefficients shows that the strongest association with age is present on the data reflecting only ECG components * indicate a "significant" effect (see Methods – Statistical Inference).



Supplementary Figure S4.

Replication FOOOF Comparison of Knee vs Fixed aperiodic modes:

Comparison between Fixed and Knee model fits using FOOOF **A)** Age was used to predict the spectral slope at rest in three different conditions (ECG components not rejected [blue], ECG components rejected [orange], ECG components only [green]). **B)** Goodness of fit was assessed using R^2 and compared between “Fixed” and “Knee” models. **C)** Comparison of the “Knee” and “Fixed” model fits shows that not fitting the knee was offering on average the better balance between goodness-of-fit and model complexity as suggested by the Bayesian Information Criterion (BIC). Error bars indicate confidence intervals.

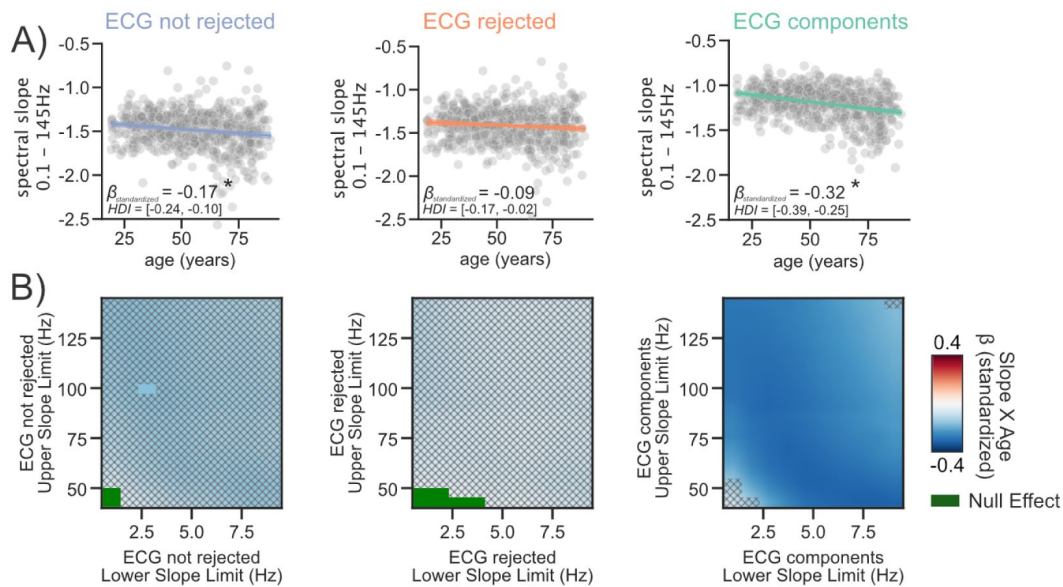


Supplementary Figure S5.

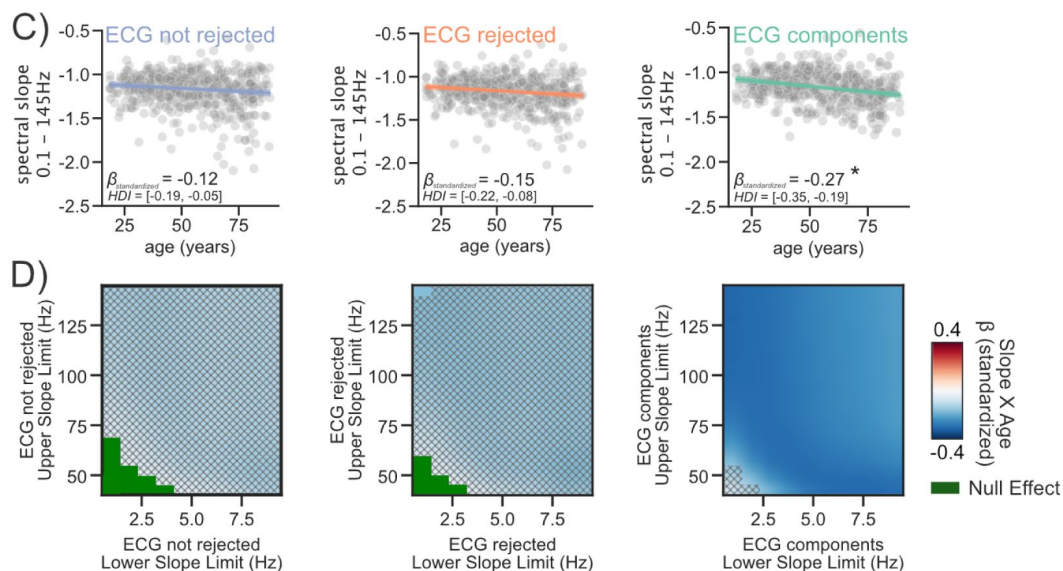
Replication Salzburg Sample:

Age-related changes in aperiodic brain activity can be explained by cardiac components **A)** Age was used to predict the spectral slope at rest in three different conditions (ECG components not rejected [blue], ECG components rejected [orange], ECG components only [green]. **B)** Age distribution in the sample routinely recorded as part of MEG measurements at the University of Salzburg. **C)** Comparison of standardized beta coefficients shows that the strongest association with age is present on the data reflecting only ECG components * indicate a “significant” effect (see Methods - Statistical Inference).

SSS Maxfilter == False



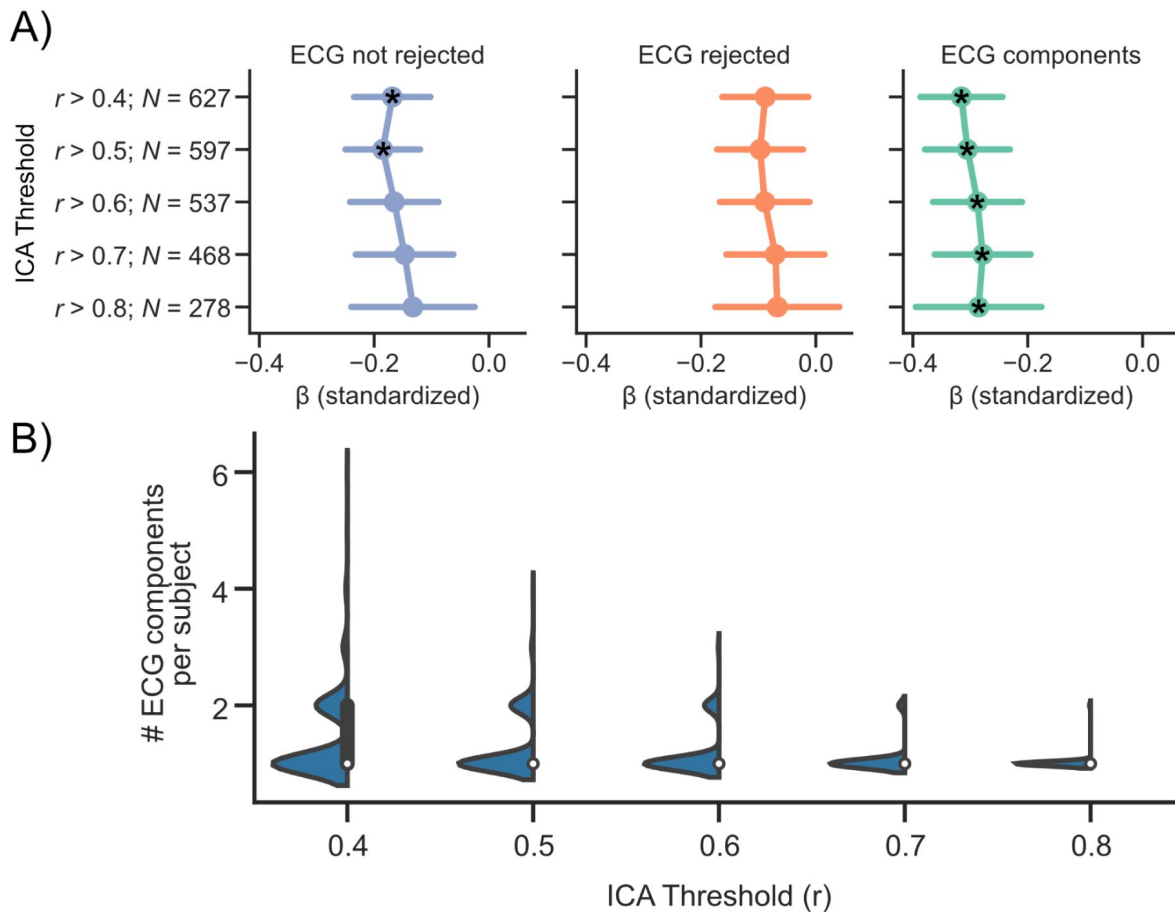
SSS Maxfilter == True



Supplementary Figure S6.

SSS Maxfilter Analysis:

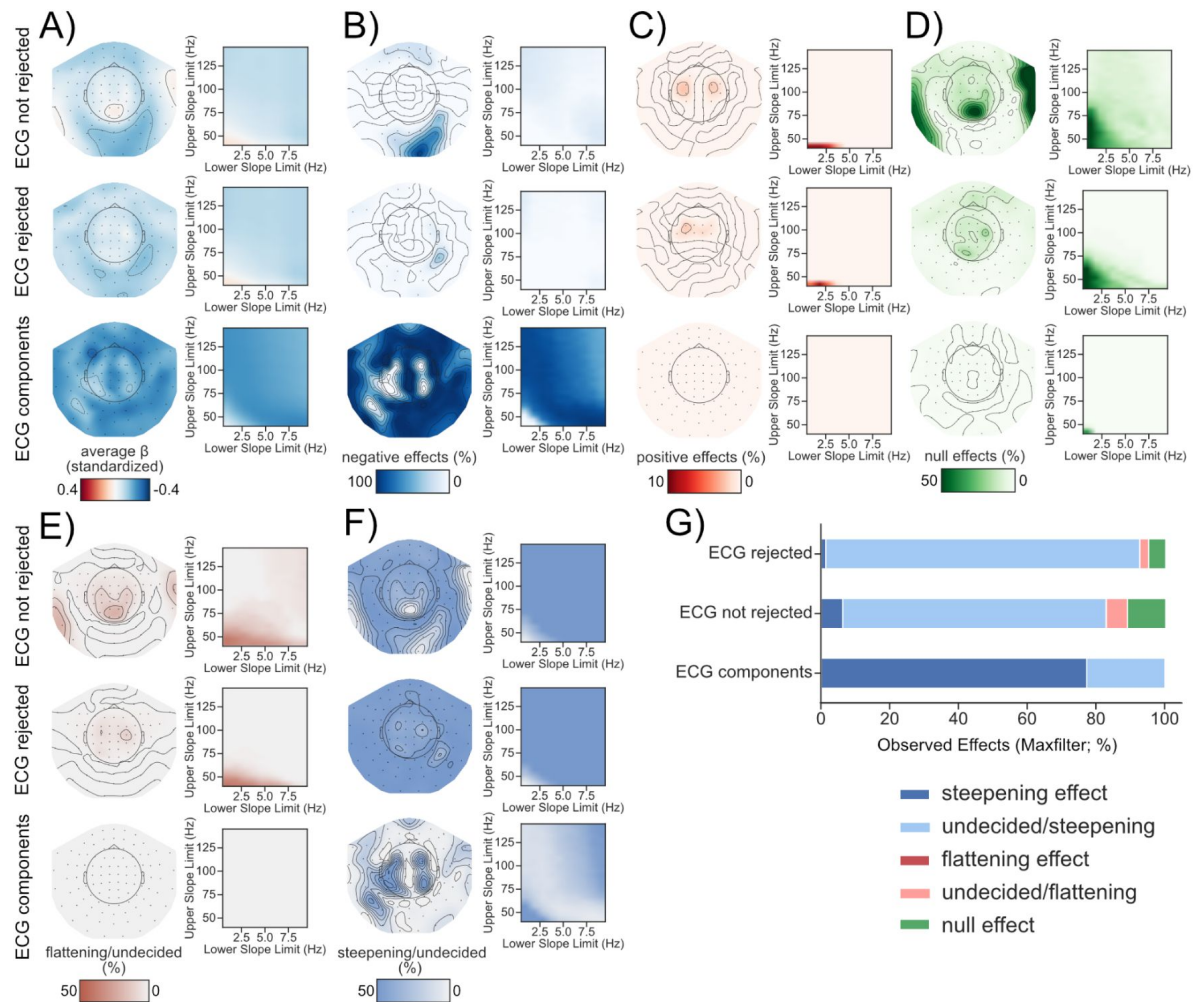
Age-related changes in aperiodic brain activity are most prominent on cardiac components irrespective of maxfiltering the data using signal space separation (SSS) or not **AC**) Age was used to predict the spectral slope (fitted at 0.1-145Hz) averaged across sensors at rest in three different conditions (ECG components not rejected [blue], ECG components rejected [orange], ECG components only [green]. **BD**) The analysis in **AC**) was repeated for different fitting ranges with lower limits starting at 0.5 Hz in 1Hz steps ranging until 10 Hz and upper limits starting at 45 Hz in 5 Hz steps ranging until 145 Hz. Significant effects, i.e. effects with credible intervals not overlapping with a region of practical equivalence (ROPE; see Methods - Statistical Inference), are highlighted in red or blue (see colorbar). Null effects, which were defined as effects with credible intervals completely within a ROPE, are highlighted in green. Results where no decision to accept or reject (see⁴⁰) an effect could be made, are masked using hatches.



Supplementary Figure S7.

ICA Thresholds:

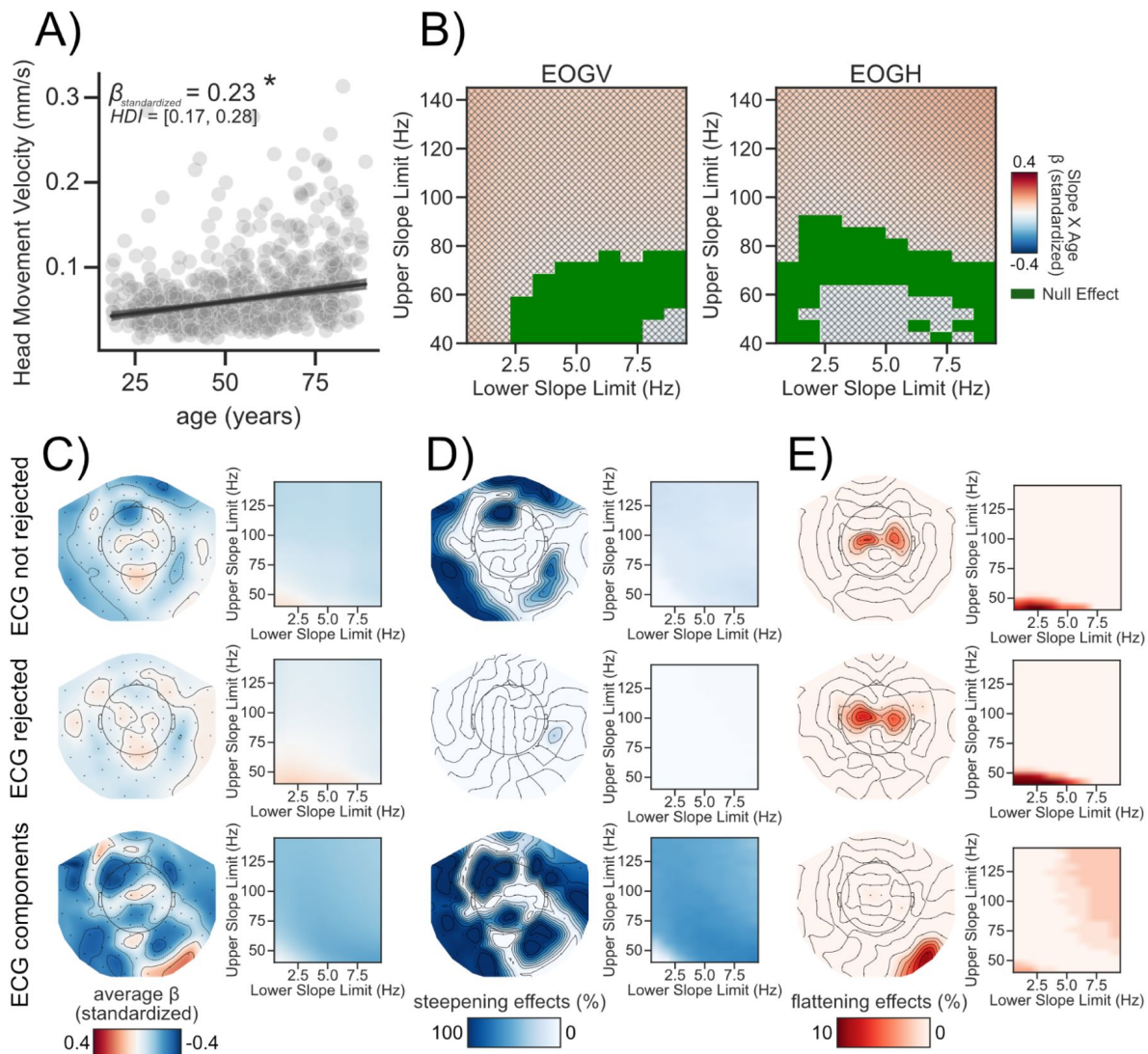
A) The relationship between age and the aperiodic slope was compared within three conditions (MEG_{ECG} not rejected [blue], MEG_{ECG} rejected [orange] and MEG_{ECG} component [green]) across different thresholds to select cardiac components from independent components of the MEG signal via ICA. Importantly, we always detected a significant relationship between age and aperiodic activity for the ECG component. While we did not always detect a significant relationship between age and aperiodic activity in MEG_{ECG} not rejected and MEG_{ECG} rejected conditions, the standardized β coefficients are heavily overlapping within each condition. Notably using a high threshold ($r > 0.8$) failed to identify cardiac components in >50% of the subjects. We therefore opted for a lower threshold of 0.4 for all related analysis in the main manuscript. Significant effects, i.e. effects with credible intervals not overlapping with a region of practical equivalence (ROPE; see Methods - Statistical Inference), are highlighted using a star. **B)** The amount of extracted ECG components per subject as a function of different ICA thresholds. On average, less than cardiac 2 components were extracted per subject irrespective of the used threshold. However, using high thresholds e.g. > 0.8 only allowed for the detection of cardiac components in less than 50% of the subjects.



Supplementary Figure S8

SSS Maxfilter Analysis:

Steepening and flattening of the spectral slope with age is dependent on the sensor location and the investigated frequency range. Age was used to predict the spectral slope at rest in three different conditions (ECG components not rejected, ECG components rejected and ECG components only) per channel across a variety of frequency ranges (see **Figure 1B**). **A)** Standardized beta coefficients either per channel averaged across all frequency ranges (left) or per frequency range (right) averaged across all channels. Age-related **B)** steepening, **C)** flattening and **D)** null effects in the spectral slope were observed and visualized in a similar manner as in **A)**. **EF)** We further show the direction of results where we didn't find enough evidence to support either a steepening, flattening or null effect. **G)** Summary of all observed findings in %.



Supplementary Figure S9.

Head Movement, EOG control analysis:

Steepening and flattening of the spectral slope with age is dependent on the recording site and the investigated frequency range, when controlling for head movements. **A)** Head movement velocity was estimated using information obtained from the cHPI measurement. **B)** Spectral slopes were obtained also for horizontal and vertical EOG electrodes on the same frequency ranges as in [Figure 4](#). Age was used to predict the spectral slope at rest in three different conditions (ECG components not rejected, ECG components rejected and ECG components only) per channel across a variety of frequency ranges. **C)** Standardized beta coefficients either per channel averaged across all frequency ranges (left) or per frequency range (right) averaged across all channels. Age-related **D)** steepening, **E)** flattening. Significant effects, i.e. effects with credible intervals not overlapping with a region of practical equivalence (ROPE; see Methods - Statistical Inference), are highlighted in red or blue (see colorbar). Null effects, which were defined as effects with credible intervals completely within a ROPE, are highlighted in green. Results where no decision to accept or reject (see [40](#)) an effect could be made, are masked using hatches.

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Reviewer #1 (Public review):**Summary:**

The present study addresses whether physiological signals influence aperiodic brain activity with a focus on age-related changes. The authors report age effects on aperiodic cardiac activity derived from ECG in low and high-frequency ranges in roughly 2300 participants from four different sites. Slopes of the ECGs were associated with common heart variability measures, which, according to the authors, shows that ECG, even at higher frequencies, conveys meaningful information. Using temporal response functions on concurrent ECG and M/EEG time series, the authors demonstrate that cardiac activity is instantaneously reflected in neural recordings, even after applying ICA analysis to remove cardiac activity. This was more strongly the case for EEG than MEG data. Finally, spectral parameterization was done in large-scale resting-state MEG and ECG data in individuals between 18 and 88 years, and age effects were tested. A steepening of spectral slopes with age was observed, particularly for ECG and, to a lesser extent, in cleaned MEG data in most frequency ranges and sensors investigated. The authors conclude that commonly observed age effects on neural aperiodic activity can mainly be explained by cardiac activity.

Strengths:

Compared to previous investigations, the authors demonstrate effects of aging on the spectral slope in the currently largest MEG dataset with equal age distribution available. Their efforts of replicating observed effects in another large MEG dataset and considering potential confounding by ocular activity, head movements, or preprocessing methods are commendable and highly valuable to the community. This study also employs a wide range of fitting ranges and two commonly used algorithms for spectral parameterization of neural and cardiac activity, hence providing a comprehensive overview of the impact of methodological choices. The authors discuss their findings in-depth and give recommendations for the separation of physiological and neural sources of aperiodic activity.

Weaknesses:

While the study's aim is well-motivated and analyses rigorously conducted, it remains vague what is reflected in the ECG at higher frequency ranges that contributed to the confounding of the age effects in the neural data. However, the authors address this issue in their discussion.

<https://doi.org/10.7554/eLife.100605.2.sa3>

Reviewer #2 (Public review):

As remains obvious from my previous reviews, I still consider this to be an important paper and that is final and publishable in its current state.

In that previous review, I revealed my identity to help reassure the authors that I was doing my best to remain unbiased because I work in this area and some of the authors' results directly impact my prior research. I was genuinely excited to see the earlier preprint version of this paper when it first appeared. I get a lot of joy out of trying to - collectively, as a field - really understand the nature of our data, and I continue to commend the authors here for pushing at the sources of aperiodic activity!

In their manuscript, Schmidt and colleagues provide a very compelling, convincing, thorough, and measured set of analyses. Previously I recommended that the push even further, and they added the current Figure 5 analysis of event-related changes in the ECG

during working memory. In my opinion this result practically warrants a separate paper its own!

The literature analysis is very clever, and expanded upon from any other prior version I've seen.

In my previous review, the broadest, most high-level comment I wanted to make was that authors are correct. We (in my lab) have tried to be measured in our approach to talking about aperiodic analyses - including adopting measuring ECG when possible now - because there are so many sources of aperiodic activity: neural, ECG, respiration, skin conductance, muscle activity, electrode impedances, room noise, electronics noise, etc. The authors discuss this all very clearly, and I commend them on that. We, as a field, should move more toward a model where we can account for all of those sources of noise together. (This was less of an action item, and more of an inclusion of a comment for the record.)

I also very much appreciate the authors' excellent commentary regarding the physiological effects that pharmacological challenges such as propofol and ketamine also have on non-neural (autonomic) functions such as ECG. Previously I also asked them to discuss the possibility that, while their manuscript focuses on aperiodic activity, it is possible that the wealth of literature regarding age-related changes in "oscillatory" activity might be driven partly by age-related changes in neural (or non-neural, ECG-related) changes in aperiodic activity. They have included a nice discussion on this, and I'm excited about the possibilities for cognitive neuroscience as we move more in this direction.

Finally, I previously asked for recommendations on how to proceed. The authors convinced me that we should care about how the ECG might impact our field potential measures, but how do I, as a relative novice, proceed. They now include three strong recommendations at the end of their manuscript that I find to be very helpful.

As was obvious from previous review, I consider this to be an important and impactful cautionary report, that is incredibly well supported by multiple thorough analyses. The authors have done an excellent job responding to all my previous comments and concerns and, in my estimation, those of the previous reviewers as well.

<https://doi.org/10.7554/eLife.100605.2.sa2>

Reviewer #3 (Public review):

Summary:

Schmidt et al., aimed to provide an extremely comprehensive demonstration of the influence cardiac electromagnetic fields have on the relationship between age and the aperiodic slope measured from electroencephalographic (EEG) and magnetoencephalographic (MEG) data.

Strengths:

Schmidt et al., used a multiverse approach to show that the cardiac influence on this relationship is considerable, by testing a wide range of different analysis parameters (including extensive testing of different frequency ranges assessed to determine the aperiodic fit), algorithms (including different artifact reduction approaches and different aperiodic fitting algorithms), and multiple large datasets to provide conclusions that are robust to the vast majority of potential experimental variations.

The study showed that across these different analytical variations, the cardiac contribution to aperiodic activity measured using EEG and MEG is considerable, and likely influences the

relationship between aperiodic activity and age to a greater extent than the influence of neural activity.

Their findings have significant implications for all future research that aims to assess aperiodic neural activity, suggesting control for the influence of cardiac fields is essential.

Weaknesses:

The authors have addressed the weaknesses of their study in their manuscript. Most alternative explanations for their results have been explored to ensure their conclusions are robust and are not explained by unexplored confounds. Minor potential weaknesses are:

(1) The number of electrodes used in the EEG analyses was on the lower side, and as such, the results do not confirm that the influence of ECG on the 1/f activity in the EEG is high even for higher density EEG montages where ICA may provide better performance at removing cardiac components (as noted by the authors). Having noted this potential weakness, I doubt the effects of cardiac activity can be completely mitigated with current methods, even in higher-density EEG montages.

(2) Head movements were used as a proxy for muscle activity. However, this may imperfectly address the potential influence of muscle activity on the slope in the EEG activity. As such, remaining muscle artifacts may have affected some of the results, particularly those that included high frequency ranges in the aperiodic estimate. Perhaps if muscle activity were left in the EEG data, it could have disrupted the ability to detect a relationship between age and 1/f slope in a way that didn't disrupt the same relationship in the cardiac data. However, I doubt this would reverse the overall conclusions given the number of converging results, including in lower frequency bands. The authors also note this potential weakness and suggest how future research might address it.

<https://doi.org/10.7554/eLife.100605.2.sa1>

Author response:

The following is the authors' response to the original reviews

eLife Assessment

Examination of (a)periodic brain activity has gained particular interest in the last few years in the neuroscience fields relating to cognition, disorders, and brain states. Using large EEG/MEG datasets from younger and older adults, the current study provides compelling evidence that age-related differences in aperiodic EEG/MEG signals can be driven by cardiac rather than brain activity. Their findings have important implications for all future research that aims to assess aperiodic neural activity, suggesting control for the influence of cardiac signals is essential.

We want to thank the editors for their assessment of our work and highlighting its importance for the understanding of aperiodic neural activity. Additionally, we want to thank the three present and four former reviewers (at a different journal) whose comments and ideas were critical in shaping this manuscript to its current form. We hope that this paper opens up many more questions that will guide us - as a field - to an improved understanding of how “cortical” and “cardiac” changes in aperiodic activity are linked and want to invite readers to engage with our work through eLife's comment function.

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Weaknesses:

While the aim of the study is well-motivated and analyses rigorously conducted, the overall structure of the manuscript, as it stands now, is partially misleading. Some of the described results are not well-embedded and lack discussion.

We want to thank the reviewer for their comments focussed on improving the overall structure of the manuscript. We agree with their suggestions that some results could be more clearly contextualized and restructured the manuscript accordingly.

Reviewer #2 (Public review):

I previously reviewed this important and timely manuscript at a previous journal where, after two rounds of review, I recommended publication. Because eLife practices an open reviewing format, I will recapitulate some of my previous comments here, for the scientific record.

In that previous review, I revealed my identity to help reassure the authors that I was doing my best to remain unbiased because I work in this area and some of the authors' results directly impact my prior research. I was genuinely excited to see the earlier preprint version of this paper when it first appeared. I get a lot of joy out of trying to - collectively, as a field - really understand the nature of our data, and I continue to commend the authors here for pushing at the sources of aperiodic activity!

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I also very much appreciate the authors' excellent commentary regarding the physiological effects that pharmacological challenges such as propofol and ketamine also have on non-neural (autonomic) functions such as ECG. Previously I also asked them to discuss the possibility that, while their manuscript focuses on aperiodic activity, it is possible that the wealth of literature regarding age-related changes in "oscillatory" activity might be driven partly by age-related changes in neural (or non-neural, ECG-related) changes in aperiodic activity. They have included a nice discussion on this, and I'm excited about the possibilities for cognitive neuroscience as we move more in this direction.

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As was obvious from previous review, I consider this to be an important and impactful cautionary report, that is incredibly well supported by multiple thorough analyses. The authors have done an excellent job responding to all my previous comments and concerns and, in my estimation, those of the previous reviewers as well.

We want to thank the reviewer for agreeing to review our manuscript again and for recapitulating on their previous comments and the progress the manuscript has made over the course of the last ~2 years. The reviewer's comments have been essential in shaping the manuscript into its current form. Their feedback has made the review process truly feel like a collaborative effort, focused on strengthening the manuscript and refining its conclusions and resulting recommendations.

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Their findings have significant implications for all future research that aims to assess aperiodic neural activity, suggesting control for the influence of cardiac fields is essential.

We want to thank the reviewer for their thorough engagement with our work and the resultant substantive amount of great ideas both mentioned in the section of Weaknesses and Authors Recommendations below. Their suggestions have sparked many ideas in us on how to move forward in better separating peripheral- from neuro-physiological signals that are likely to greatly influence our future attempts to better extract both cardiac and muscle activity from M/EEG recordings. So we want to thank them for their input, time and effort!

Weaknesses:

Figure 4I: The regressions explained here seem to contain a very large number of potential predictors. Based on the way it is currently written, I'm assuming it includes all sensors for both the ECG component and ECG rejected conditions?

I'm not sure about the logic of taking a complete signal, decomposing it with ICA to separate out the ECG and non-ECG signals, then including these latent contributions to the full signal back into the same regression model. It seems that there could be some circularity or redundancy in doing so. Can the authors provide a justification for why this is a valid approach?

After observing significant effects both in the $MEG_{ECG \text{ component}}$ and $MEG_{ECG \text{ rejected}}$ conditions in similar frequency bands we wanted to understand whether or not these age-related changes are statistically independent. To test this we added both variables as predictors in a regression model (thereby accounting for the influence of the other in relation to age). The regression models we performed were therefore actually not very complex. They were built using only two predictors, namely the data (in a specific frequency range) averaged over channels on which we noticed significant effects in the ECG rejected and ECG components data respectively (Wilkinson notation: $age \sim 1 + ECG \text{ rejected} + ECG \text{ components}$). This was also described in the results section stating that: “To see if $MEG_{ECG \text{ rejected}}$ and $MEG_{ECG \text{ component}}$ explain unique variance in aging at frequency ranges where we noticed shared effects, we averaged the spectral slope across significant channels and calculated a multiple regression model with $MEG_{ECG \text{ component}}$ and $MEG_{ECG \text{ rejected}}$ as predictors for age (to statistically control for the effect of $MEG_{ECG \text{ component}}$ s and $MEG_{ECG \text{ rejected}}$ on age). This analysis was performed to understand whether the observed shared age-related effects ($MEG_{ECG \text{ rejected}}$ and $MEG_{ECG \text{ component}}$) are in(dependent).”

We hope this explanation solves the previous misunderstanding.

I'm not sure whether there is good evidence or rationale to support the statement in the discussion that the presence of the ECG signal in reference electrodes makes it more

difficult to isolate independent ECG components. The ICA algorithm will still function to detect common voltage shifts from the ECG as statistically independent from other voltage shifts, even if they're spread across all electrodes due to the referencing montage. I would suggest there are other reasons why the ICA might lead to imperfect separation of the ECG component (assumption of the same number of source components as sensors, non-Gaussian assumption, assumption of independence of source activities).

The inclusion of only 32 channels in the EEG data might also have reduced the performance of ICA, increasing the chances of imperfect component separation and the mixing of cardiac artifacts into the neural components, whereas the higher number of sensors in the MEG data would enable better component separation. This could explain the difference between EEG and MEG in the ability to clean the ECG artifact (and perhaps higher-density EEG recordings would not show the same issue).

The reviewer is making a good argument suggesting that our initial assumption that the presence of cardiac activity on the reference electrode influences the performance of the ICA may be wrong. After rereading and rethinking upon the matter we think that the reviewer is correct and that their assumptions for why the ECG signal was not so easily separable from our EEG recordings are more plausible and better grounded in the literature than our initial suggestion. We therefore now highlight their view as a main reason for why the ECG rejection was more challenging in EEG data. However, we also note that understanding the exact reason probably ends up being an empirical question that demands further research stating that:

“Difficulties in removing ECG related components from EEG signals via ICA might be attributable to various reasons such as the number of available sensors or assumptions related to the non-gaussianity of the underlying sources. Further understanding of this matter is highly important given that ICA is the most widely used procedure to separate neural from peripheral physiological sources.”

In addition to the inability to effectively clean the ECG artifact from EEG data, ICA and other component subtraction methods have also all been shown to distort neural activity in periods that aren't affected by the artifact due to the ubiquitous issue of imperfect component separation (<https://doi.org/10.1101/2024.06.06.597688>). As such, component subtraction-based (as well as regression-based) removal of the cardiac artifact might also distort the neural contributions to the aperiodic signal, so even methods to adequately address the cardiac artifact might not solve the problem explained in the study. This poses an additional potential confound to the "M/EEG without ECG" conditions.

The reviewer is correct in stating that, if an “artifactual” signal is not always present but appears and disappears (like e.g. eye-blinks) neural activity may be distorted in periods where the “artifactual” signal is absent. However, while this plausibly presents a problem for ocular activity, there is no obvious reason to believe that this applies to cardiac activity. While the ECG signal is non-stationary in nature, it is remarkably more stable than eye-movements in the healthy populations we analyzed (especially at rest). Therefore, the presence of the cardiac “artifact” was consistently present across the entirety of the MEG recordings we visually inspected.

Literature Analysis, Page 23: was there a method applied to address studies that report reducing artifacts in general, but are not specific to a single type of artifact? For example, there are automated methods for cleaning EEG data that use ICLabel (a machine learning algorithm) to delete "artifact" components. Within these studies, the cardiac artifact will not be mentioned specifically, but is included under "artifacts".

The literature analysis was largely performed automatically and solely focussed on ECG related activity as described in the methods section under Literature Analysis, if no ECG related terms were used in the context of artifact rejection a study was flagged as not having removed cardiac activity. This could have been indeed better highlighted by us and we apologize for the oversight on our behalf. We now additionally link to these details stating that:

“However, an analysis of openly accessible M/EEG articles ($N_{Articles}=279$; see Methods - Literature Analysis for further details) that investigate aperiodic activity revealed that only 17.1% of EEG studies explicitly mention that cardiac activity was removed and only 16.5% measure ECG (45.9% of MEG studies removed cardiac activity and 31.1% of MEG studies mention that ECG was measured; see Figure 1EF).”

The reviewer makes a fair point that there is some uncertainty here and our results probably present a lower bound of ECG handling in M/EEG research as, when I manually rechecked the studies that were not initially flagged in studies it was often solely mentioned that “artifacts” were rejected. However, this information seemed too ambiguous to assume that cardiac activity was in fact accounted for. However, again this could have been mentioned more clearly in writing and we apologize for this oversight. Now this is included as part of the methods section *Literature Analysis* stating that:

“All valid word contexts were then manually inspected by scanning the respective word context to ensure that the removal of “artifacts” was related specifically to cardiac and not e.g. ocular activity or the rejection of artifacts in general (without specifying which “artifactual” source was rejected in which case the manuscript was marked as invalid). This means that the results of our literature analysis likely present a lower bound for the rejection of cardiac activity in the M/EEG literature investigating aperiodic activity.”

Statistical inferences, page 23: as far as I can tell, no methods to control for multiple comparisons were implemented. Many of the statistical comparisons were not independent (or even overlapped with similar analyses in the full analysis space to a large extent), so I wouldn't expect strong multiple comparison controls. But addressing this point to some extent would be useful (or clarifying how it has already been addressed if I've missed something).

In the present study we tried to minimize the risk of type 1 errors by several means, such as A) weakly informative priors, B) robust regression models and C) by specifying a region of practical equivalence (ROPE, see Methods Statistical Inference for further Information) to define meaningful effects.

Weakly informative priors can lower the risk of type 1 errors arising from multiple testing by shrinking parameter estimates towards zero (see e.g. Lemoine, 2019). Robust regression models use a Student T distribution to describe the distribution of the data. This distribution features heavier tails, meaning it allocates more probability to extreme values, which in turn minimizes the influence of outliers. The ROPE criterion ensures that only effects exceeding a negligible size are considered meaningful, representing a strict and conservative approach to interpreting our findings (see Kruschke 2018, Cohen, 1988).

Furthermore, and more generally we do not selectively report “significant” effects in the situations in which multiple analyses were conducted on the same family of data (e.g. Figure 2 & 4). Instead we provide joint inference across several plausible analysis options (akin to a specification curve analysis, Simonsohn, Simmons & Nelson 2020) to provide other researchers with an overview of how different analysis choices impact the association between cardiac and neural aperiodic activity.

Lemoine, N. P. (2019). Moving beyond noninformative priors: why and how to choose weakly informative priors in Bayesian analyses. *Oikos*, 128(7), 912-928.

Simonsohn, U., Simmons, J. P., & Nelson, L. D. (2020). Specification curve analysis. *Nature Human Behaviour*, 4(11), 1208-1214.

Methods:

Applying ICA components from 1Hz high pass filtered data back to the 0.1Hz filtered data leads to worse artifact cleaning performance, as the contribution of the artifact in the 0.1Hz to 1Hz frequency band is not addressed (see Bailey, N. W., Hill, A. T., Biabani, M., Murphy, O. W., Rogasch, N. C., McQueen, B., ... & Fitzgerald, P. B. (2023). RELAX part 2: A fully automated EEG data cleaning algorithm that is applicable to Event-Related-Potentials. Clinical Neurophysiology, result reported in the supplementary materials). This might explain some of the lower frequency slope results (which include a lower frequency limit <1Hz) in the EEG data - the EEG cleaning method is just not addressing the cardiac artifact in that frequency range (although it certainly wouldn't explain all of the results).

We want to thank the reviewer for suggesting this interesting paper, showing that lower high-pass filters may be preferable to the more commonly used >1Hz high-pass filters for detection of ICA components that largely contain peripheral physiological activity. However, the results presented by Bailey et al. contradict the more commonly reported findings by other researchers that >1Hz high-pass filter is actually preferable (e.g. Winkler et al. 2015; Dimigen, 2020 or Klug & Gramann, 2021) and recommendations in widely used packages for M/EEG analysis (e.g. <https://mne.tools/1.8/generated/mne.preprocessing.ICA.html>). Yet, the fact that there seems to be a discrepancy suggests that further research is needed to better understand which type of high-pass filtering is preferable in which situation. Furthermore, it is notable that all the findings for high-pass filtering in ICA component detection and removal that we are aware of relate to ocular activity. Given that ocular and cardiac activity have very different temporal and spectral patterns it is probably worth further investigating whether the classic 1Hz high-pass filter is really also the best option for the detection and removal of cardiac activity. However, in our opinion this requires a dedicated investigation on its own..

We therefore highlight this now in our manuscript stating that:

“Additionally, it is worth noting that the effectiveness of an ICA crucially depends on the quality of the extracted components(63,64) and even widely suggested settings e.g. high-pass filtering at 1Hz before fitting an ICA may not be universally applicable (see supplementary material of (64)).

Winkler, S. Debener, K. -R. Müller and M. Tangermann, "On the influence of high-pass filtering on ICA-based artifact reduction in EEG-ERP," 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Milan, Italy, 2015, pp. 4101-4105, doi: 10.1109/EMBC.2015.7319296.

Dimigen, O. (2020). Optimizing the ICA-based removal of ocular EEG artifacts from free viewing experiments. *NeuroImage*, 207, 116117.

Klug, M., & Gramann, K. (2021). Identifying key factors for improving ICA-based decomposition of EEG data in mobile and stationary experiments. *European Journal of Neuroscience*, 54(12), 8406-8420.

It looks like no methods were implemented to address muscle artifacts. These can affect the slope of EEG activity at higher frequencies. Perhaps the Riemannian Potato addressed these artifacts, but I suspect it wouldn't eliminate all muscle activity. As such, I

would be concerned that remaining muscle artifacts affected some of the results, particularly those that included high frequency ranges in the aperiodic estimate. Perhaps if muscle activity were left in the EEG data, it could have disrupted the ability to detect a relationship between age and 1/f slope in a way that didn't disrupt the same relationship in the cardiac data (although I suspect it wouldn't reverse the overall conclusions given the number of converging results including in lower frequency bands). Is there a quick validity analysis the authors can implement to confirm muscle artifacts haven't negatively affected their results?

*I note that an analysis of head movement in the MEG is provided on page 32, but it would be more robust to show that removing ICA components reflecting muscle doesn't change the results. The results/conclusions of the following study might be useful for objectively detecting probable muscle artifact components: Fitzgibbon, S. P., DeLosAngeles, D., Lewis, T. W., Powers, D. M. W., Grummett, T. S., Whitham, E. M., ... & Pope, K. J. (2016). Automatic determination of EMG-contaminated components and validation of independent component analysis using EEG during pharmacologic paralysis. *Clinical neurophysiology*, 127(3), 1781-1793.*

We thank the reviewer for their suggestion. Muscle activity can indeed be a potential concern, for the estimation of the spectral slope. This is precisely why we used head movements (as also noted by the reviewer) as a proxy for muscle activity. We also agree with the reviewer that this is not a perfect estimate. Additionally, also the riemannian potato would probably only capture epochs that contain transient, but not persistent patterns of muscle activity.

The paper recommended by the reviewer contains a clever approach of using the steepness of the spectral slope (or lack thereof) as an indicator whether or not an independent component (IC) is driven by muscle activity. In order to determine an optimal threshold Fitzgibbon et al. compared paralyzed to temporarily non paralyzed subjects. They determined an expected “EMG-free” threshold for their spectral slope on paralyzed subjects and used this as a benchmark to detect IC's that were contaminated by muscle activity in non paralyzed subjects.

This is a great idea, but unfortunately would go way beyond what we are able to sensibly estimate with our data for the following reasons. The authors estimated their optimal threshold on paralyzed subjects for EEG data and show that this is a feasible threshold to be applied across different recordings. So for EEG data it might be feasible, at least as a first shot, to use their threshold on our data. However, we are measuring MEG and as alluded to in our discussion section under “Differences in aperiodic activity between magnetic and electric field recordings” the spectral slope differs greatly between MEG and EEG recordings for non-trivial reasons. Furthermore, the spectral slope even seems to also differ across different MEG devices. We noticed this when we initially tried to pool the data recorded in Salzburg with the Cambridge dataset. This means we would need to do a complete validation of this procedure for the MEG data recorded in Cambridge and in Salzburg, which is not feasible considering that we A) don't have direct access to one of the recording sites and B) would even if we had access face substantial hurdles to get ethical approval for the experiment performed by Fitzgibbon et al..

However, we think the approach brought forward by Fitzgibbon and colleagues is a clever way to remove muscle activity from EEG recordings, whenever EMG was not directly recorded. We therefore suggested in the Discussion section that ideally also EMG should be recorded stating that:

“It is worth noting that, apart from cardiac activity, muscle activity can also be captured in (non-)invasive recordings and may drastically influence measures of the spectral slope(72). To ensure that persistent muscle activity does not bias our results we used changes in head

movement velocity as a control analysis (see Supplementary Figure S9). However, it should be noted that this is only a proxy for the presence of persistent muscle activity. Ideally, studies investigating aperiodic activity should also be complemented by measurements of EMG. Whenever such measurements are not available creative approaches that use the steepness of the spectral slope (or the lack thereof) as an indicator to detect whether or not e.g. an independent component is driven by muscle activity are promising(72,73). However, these approaches may require further validation to determine how well myographic aperiodic thresholds are transferable across the wide variety of different M/EEG devices.”

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) As outlined above, I recommend rephrasing the last section of the introduction to briefly summarize/introduce all main analysis steps undertaken in the study and why these were done (for example, it is only mentioned that the Cam-CAN dataset was used to study the impact of cardiac on MEG activity although the author used a variety of different datasets). Similarly, I am missing an overview of all main findings in the context of the study goals in the discussion. I believe clarifying the structure of the paper would not only provide a red thread to the reader but also highlight the efforts/strength of the study as described above.

This is a good call! As suggested by the reviewer we now try to give a clearer overview of what was investigated why. We do that both at the end of the introduction stating that: “Using the publicly available Cam-CAN dataset(28,29), we find that the aperiodic signal measured using M/EEG originates from multiple physiological sources. In particular, significant portions of age-related changes in aperiodic activity –normally attributed to neural processes– can be better explained by cardiac activity. This observation holds across a wide range of processing options and control analyses (see Supplementary S1), and was replicable on a separate MEG dataset. However, the extent to which cardiac activity accounts for age-related changes in aperiodic activity varies with the investigated frequency range and recording site. Importantly, in some frequency ranges and sensor locations, age-related changes in neural aperiodic activity still prevail. But does the influence of cardiac activity on the aperiodic spectrum extend beyond age? In a preliminary analysis, we demonstrate that working memory load modulates the aperiodic spectrum of “pure” ECG recordings. The direction of this working memory effect mirrors previous findings on EEG data(5) suggesting that the impact of cardiac activity goes well beyond aging. In sum, our results highlight the complexity of aperiodic activity while cautioning against interpreting it as solely “neural” without considering physiological influences.”

and at the beginning of the discussion section:

“Difficulties in removing ECG related components from EEG signals via ICA might be attributable to various reasons such as the number of available sensors or assumptions related to the non-gaussianity of the underlying sources. Further understanding of this matter is highly important given that ICA is the most widely used procedure to separate neural from peripheral physiological sources (see Figure 1EF). Additionally, it is worth noting that the effectiveness of an ICA crucially depends on the quality of the extracted components(63,64) and even widely suggested settings e.g. high-pass filtering at 1Hz before fitting an ICA may not be universally applicable (see supplementary material of (64)). “

(2) I found it interesting that the spectral slopes of ECG activity at higher frequency ranges (> 10 Hz) seem mostly related to HRV measures such as fractal and time domain indices and less so with frequency-domain indices. Do the authors have an explanation

for why this is the case? Also, the analysis of the HRV measures and their association with aperiodic ECG activity is not explained in any of the method sections.

We apologize for the oversight in not mentioning the HRV analysis in more detail in our methods section. We added a subsection to the Methods section entitled ECG Processing - Heart rate variability analysis to further describe the HRV analyses.

“ECG Processing - Heart rate variability analysis

Heart rate variability (HRV) was computed using the NeuroKit2 toolbox, a high level tool for the analysis of physiological signals. First, the raw electrocardiogram (ECG) data were preprocessed, by highpass filtering the signal at 0.5Hz using an infinite impulse response (IIR) butterworth filter(order=5) and by smoothing the signal with a moving average kernel with the width of one period of 50Hz to remove the powerline noise (default settings of neurokit.ecg.ecg_clean). Afterwards, QRS complexes were detected based on the steepness of the absolute gradient of the ECG signal. Subsequently, R-Peaks were detected as local maxima in the QRS complexes (default settings of neurokit.ecg.ecg_peaks; see (98) for a validation of the algorithm). From the cleaned R-R intervals, 90 HRV indices were derived, encompassing time-domain, frequency-domain, and non-linear measures. Time-domain indices included standard metrics such as the mean and standard deviation of the normalized R-R intervals, the root mean square of successive differences, and other statistical descriptors of interbeat interval variability. Frequency-domain analyses were performed using power spectral density estimation, yielding for instance low frequency (0.04-0.15Hz) and high frequency (0.15-0.4Hz) power components. Additionally, non-linear dynamics were characterized through measures such as sample entropy, detrended fluctuation analysis and various Poincaré plot descriptors. All these measures were then related to the slopes of the low frequency (0.25 – 20 Hz) and high frequency (10 – 145 Hz) aperiodic spectrum of the raw ECG.”

With regards to association of the ECG’s spectral slopes at high frequencies and frequency domain indices of heart rate variability. Common frequency domain indices of heart rate variability fall in the range of 0.01-4Hz. Which probably explains why we didn’t notice any association at higher frequency ranges (>10Hz).

This is also stated in the related part of the results section:

“In the higher frequency ranges (10 - 145 Hz) spectral slopes were most consistently related to fractal and time domain indices of heart rate variability, but not so much to frequency-domain indices assessing spectral power in frequency ranges < 0.4 Hz.”

(3) Related to the previous point - what is being reflected in the ECG at higher frequency ranges, with regard to biological mechanisms? Results are being mentioned, but not further discussed. However, this point seems crucial because the age effects across the four datasets differ between low and high-frequency slope limits (Figure 2C).

This is a great question that definitely also requires further attention and investigation in general (see also Tereshchenko & Josephson, 2015). We investigated the change of the slope across frequency ranges that are typically captured in common ECG setups for adults (0.05 - 150Hz, Tereshchenko & Josephson, 2015; Kusayama, Wong, Liu et al. 2020). While most of the physiological significant spectral information of an ECG recording rests between 1-50Hz (Clifford & Azuaje, 2006), meaningful information can be extracted at much higher frequencies. For instance, ventricular late potentials have a broader frequency band (~40-250Hz) that falls straight in our spectral analysis window. However, that’s not all, as further meaningful information can be extracted at even higher frequencies (>100Hz). Yet, the exact physiological mechanisms underlying so-called high-frequency QRS remain unclear (HF-QRS; see Tereshchenko & Josephson, 2015; Qiu et al. 2024 for a review discussing possible

mechanisms). Yet, at the same time the HF-QRS seems to be highly informative for the early detection of myocardial ischemia and other cardiac abnormalities that may not yet be evident in the standard frequency range (Schlegel et al. 2004; Qiu et al. 2024). All optimism aside, it is also worth noting that ECG recordings at higher frequencies can capture skeletal muscle activity with an overlapping frequency range up to 400Hz (Kusayama, Wong, Liu et al. 2020). We highlight all of this now when introducing this analysis in the results sections as outstanding research question stating that:

“However, substantially less is known about aperiodic activity above 0.4Hz in the ECG. Yet, common ECG setups for adults capture activity at a broad bandwidth of 0.05 - 150Hz(33,34).

Importantly, a lot of the physiological meaningful spectral information rests between 1-50Hz(35), similarly to M/EEG recordings. Furthermore, meaningful information can be extracted at much higher frequencies. For instance, ventricular late potentials have a broader frequency band (~40-250Hz(35)). However, that’s not all, as further meaningful information can be extracted at even higher frequencies (>100Hz). For instance, the so-called high-frequency QRS seems to be highly informative for the early detection of myocardial ischemia and other cardiac abnormalities that may not yet be evident in the standard frequency range(36,37). Yet, the exact physiological mechanisms underlying the high-frequency QRS remain unclear (see (37) for a review discussing possible mechanisms).”

Tereshchenko, L. G., & Josephson, M. E. (2015). Frequency content and characteristics of ventricular conduction. *Journal of electrocardiology*, 48(6), 933-937.

Kusayama, T., Wong, J., Liu, X. et al. Simultaneous noninvasive recording of electrocardiogram and skin sympathetic nerve activity (neuECG). *Nat Protoc* 15, 1853–1877 (2020). <https://doi.org/10.1038/s41596-020-0316-6>

Clifford, G. D., & Azuaje, F. (2006). *Advanced methods and tools for ECG data analysis* (Vol. 10). P. McSharry (Ed.). Boston: Artech house.

Qiu, S., Liu, T., Zhan, Z., Li, X., Liu, X., Xin, X., ... & Xiu, J. (2024). Revisiting the diagnostic and prognostic significance of high-frequency QRS analysis in cardiovascular diseases: a comprehensive review. *Postgraduate Medical Journal*, qgae064.

Schlegel, T. T., Kulecz, W. B., DePalma, J. L., Feiveson, A. H., Wilson, J. S., Rahman, M. A., & Bungo, M. W. (2004, March). Real-time 12-lead high-frequency QRS electrocardiography for enhanced detection of myocardial ischemia and coronary artery disease. In *Mayo Clinic Proceedings* (Vol. 79, No. 3, pp. 339-350). Elsevier.

(4) Page 10: At first glance, it is not quite clear what is meant by "processing option" in the text. Please clarify.

Thank you for catching this! Upon re-reading this is indeed a bit oblivious. We now swapped “processing options” with “slope fits” to make it clearer that we are talking about the percentage of effects based on the different slope fits.

(5) The authors mention previous findings on age effects on neural 1/f activity (References Nr 5,8,27,39) that seem contrary to their own findings such as e.g., the mostly steepening of the slopes with age. Also, the authors discuss thoroughly why spectral slopes derived from MEG signals may differ from EEG signals. I encourage the authors to have a closer look at these studies and elaborate a bit more on why these studies differ in their conclusions on the age effects. For example, Tröndle et al. (2022, Ref. 39) investigated neural activity in children and young adults, hence, focused on brain maturation, whereas the CamCAN set only considers the adult lifespan. In a similar vein,

others report age effects on 1/f activity in much smaller samples as reported here (e.g., Voytek et al., 2015).

I believe taking these points into account by briefly discussing them, would strengthen the authors' claims and provide a more fine-grained perspective on aging effects on 1/f.

The reviewer is making a very important point. As age-related differences in (neuro-)physiological activity are not necessarily strictly comparable and entirely linear across different age-cohorts (e.g. age-related changes in alpha center frequency). We therefore, added the suggested discussion points to the discussion section.

“Differences in electric and magnetic field recordings aside, aperiodic activity may not change strictly linearly as we are ageing and studies looking at younger age groups (e.g. <22; (44) may capture different aspects of aging (e.g. brain maturation), than those looking at older subjects (>18 years; our sample). A recent report even shows some first evidence of an interesting putatively non-linear relationship with age in the sensorimotor cortex for resting recordings(59)”

(6) The analysis of the working memory paradigm as described in the outlook-section of the discussion comes as a bit of a surprise as it has not been introduced before. If the authors want to convey with this study that, in general, aperiodic neural activity could be influenced by aperiodic cardiac activity, I recommend introducing this analysis and the results earlier in the manuscript than only in the discussion to strengthen their message.

The reviewer is correct. This analysis really comes a bit out of the blue. However, this was also exactly the intention for placing this analysis in the discussion. As the reviewer correctly noted, the aim was to suggest “that, in general, aperiodic neural activity could be influenced by aperiodic cardiac activity”. We placed this outlook directly after the discussion of “(neuro-)physiological origins of aperiodic activity”, where we highlight the potential challenges of interpreting drug induced changes to M/EEG recordings. So the aim was to get the reader to think about whether age is the only feature affected by cardiac activity and then directly present some evidence that this might go beyond age.

However, we have been rethinking this approach based on the reviewers comments and moved that paragraph to the end of the results section accordingly and introduce it already at the end of the introduction stating that:

“But does the influence of cardiac activity on the aperiodic spectrum extend beyond age? In a preliminary analysis, we demonstrate that working memory load modulates the aperiodic spectrum of “pure” ECG recordings. The direction of this working memory effect mirrors previous findings on EEG data(5) suggesting that the impact of cardiac activity goes well beyond aging.”

(7) The font in Figure 2 is a bit hard to read (especially in D). I recommend increasing the font sizes where necessary for better readability.

We agree with the Reviewer and increased the font sizes accordingly.

(8) Text in the discussion: Figure 3B on page 10 => shouldn't it be Figure 4?

Thank you for catching this oversight. We have now corrected this mistake.

(9) In the third section on page 10, the Figure labels seem to be confused. For example, Figure 4 E is supposed to show "steepening effects", which should be Figure 4B I believe.

Please check the figure labels in this section to avoid confusion.

Thank you for catching this oversight. We have now corrected this mistake.

| (10) Figure Legend 4 I), please check the figure labels in the text

Thank you for catching this oversight. We have now corrected this mistake.

Reviewer #3 (Recommendations for the authors):

I have a number of suggestions for improving the manuscript, which I have divided by section in the following:

ABSTRACT:

I would suggest re-writing the first sentences to make it easier to read for non-expert readers: "The power of electrophysiologically measured cortical activity decays with an approximately $1/f^X$ function. The slope of this decay (i.e. the spectral exponent, X) is modulated..."

Thank you for the suggestion. We adjusted the sentence as suggested to make it easier for less technical readers to understand that "X" refers to the exponent.

| *Including the age range that was studied in the abstract could be informative.*

Done as suggested.

As an optional recommendation, I think it would increase the impact of the article if the authors note in the abstract that the current most commonly applied cardiac artifact reduction approaches don't resolve the issue for EEG data, likely due to an imperfect ability to separate the cardiac artifact from the neural activity with independent component analysis. This would highlight to the reader that they can't just expect to address these concerns by cleaning their data with typical cleaning methods.

I think it would also be useful to convey in the abstract just how comprehensive the included analyses were (in terms of artifact reduction methods tested, different aperiodic algorithms and frequency ranges, and both MEG and EEG). Doing so would let the reader know just how robust the conclusions are likely to be.

This is a brilliant idea! As suggested we added a sentence highlighting that simply performing an ICA may not be sufficient to separate cardiac contributions to M/EEG recordings and refer to the comprehensiveness of the performed analyses.

INTRODUCTION:

I would suggest re-writing the following sentence for readability: "In the past, aperiodic neural activity, other than periodic neural activity (local peaks that rise above the "power-law" distribution), was often treated as noise and simply removed from the signal"

To something like: "In the past, aperiodic neural activity was often treated as noise and simply removed from the signal e.g. via pre-whitening, so that analyses could focus on periodic neural activity (local peaks that rise above the "power-law" distribution, which are typically thought to reflect neural oscillations).

We are happy to follow that suggestion.

Page 3: please provide the number of articles that were included in the examination of the percentage that remove cardiac activity, and note whether the included articles could be considered a comprehensive or nearly comprehensive list, or just a representative sample.

We stated the exact number of articles in the methods section under *Literature Analysis*. However, we added it to the Introduction on page 3 as suggested by the reviewer. The selection of articles was done automatically, dependent on a list of pre-specified terms and exclusively focussed on articles that had terms related to aperiodic activity in their title (see *Literature Analysis*). Therefore, I would personally be hesitant in calling it a comprehensive or nearly comprehensive list of the general M/EEG literature as the analysis of aperiodic activity is still relatively niche compared to the more commonly investigated evoked potentials or oscillations. I think whether or not a reader perceives our analysis as comprehensive should be up to them to decide and does not reflect something I want to impose on them. This is exacerbated by the fact that the analysis of neural aperiodic activity has rapidly gained traction over the last years (see Figure 1D orange) and the literature analysis was performed almost 2 years ago and therefore, in my eyes, only represents a glimpse in the rapidly evolving field related to the analysis of aperiodic activity.

Figure 1E-F: It's not completely clear that the "Cleaning Methods" part of the figure indicates just methods to clean the cardiac artifact (rather than any artifact). It also seems that ~40% of EEG studies do not apply any cleaning methods even from within the studies that do clean the cardiac artifact (if I've read the details correctly). This seems unlikely. Perhaps there should be a bar for "other methods", or "unspecified"? Having said that, I'm quite familiar with the EEG artifact reduction literature, and I would be very surprised if ~40% of studies cleaned the cardiac artifact using a different method to the methods listed in the bar graph, so I'm wondering if I've misunderstood the figure, or whether the data capture is incomplete / inaccurate (even though the conclusion that ICA is the most common method is almost certainly accurate).

The cleaning is indeed only focussed on cardiac activity specifically. This was however also mentioned in the caption of Figure 1: “We were further interested in determining which artifact rejection approaches were most commonly used to remove cardiac activity, such as independent component analysis (ICA(22)), singular value decomposition (SVD(23)), signal space separation (SSS(24)), signal space projections (SSP(25)) and denoising source separation (DSS(26)).” and in the methods section under *Literature Analysis*. However, we adjusted figure 1EF to make it more obvious that the described cleaning methods were only related to the ECG. Aside from using blind source separation techniques such as ICA a good amount of studies mentioned that they cleaned their data based on visual inspection (which was not further considered). Furthermore, it has to be noted that only studies were marked as having separated cardiac from neural activity, when this was mentioned explicitly.

RESULTS:

Page 6: I would delete the "from a neurophysiological perspective" clause, which makes the sentence more difficult to read and isn't so accurate (frequencies 13-25Hz would probably more commonly be considered mid-range rather than low or high). Additionally, both frequency ranges include 15Hz, but the next sentence states that the ranges were selected to avoid the knee at 15Hz, which seems to be a contradiction. Could the authors explain in more detail how the split addresses the 15Hz knee?

We removed the “from a neurophysiological perspective” clause as suggested. With regards to the “knee” at ~15Hz I would like to defer the reviewer to Supplementary Figure S1. The

Knee Frequency varies substantially across subjects so splitting the data at only 1 exact Frequency did not seem appropriate. Additionally, we found only spurious significant age-related variations in Knee Frequency (i.e. only one out of the 4 datasets; not shown).

Furthermore, we wanted to better connect our findings to our MEG results in Figure 4 and also give the readers a holistic overview of how different frequency ranges in the aperiodic ECG would be affected by age. So to fulfill all of these objectives we decided to fit slopes with respective upper/lower bounds around a range of 5Hz above and below the average 15Hz Knee Frequency across datasets.

The later parts of this same paragraph refer to a vast amount of different frequency ranges, but only the "low" and "high" frequency ranges were previously mentioned. Perhaps the explanation could be expanded to note that multiple lower and upper bounds were tested within each of these low and high frequency windows?

This is a good catch we adjusted the sentence as suggested. We now write: “.. slopes were fitted individually to each subject's power spectrum in several lower (0.25 – 20 Hz) and higher (10-145 Hz) frequency ranges.”

The following two sentences seem to contradict each other: "Overall, spectral slopes in lower frequency ranges were more consistently related to heart rate variability indices(> 39.4% percent of all investigated indices)" and: "In the lower frequency range (0.25 - 20Hz), spectral slopes were consistently related to most measures of heart rate variability; i.e. significant effects were detected in all 4 datasets (see Figure 2D)." (39.4% is not "most").

The reviewer is correct in stating that 39.4% is not most. However, the 39.4% is the lowest bound and only refers to 1 dataset. In the other 3 datasets the percentage of effects was above 64% which can be categorized as “most” i.e. above 50%. We agree that this was a bit ambiguous in the sentence so we added the other percentages as well as a reference to Figure 2D to make this point clearer.

Figure 2D: it isn't clear what the percentages in the semi-circles reflect, nor why some semi-circles are more full circles while others are only quarter circles.

The percentages in the semi-circles reflect the amount of effects (marked in red) and null effects (marked in green) per dataset, when viewed as average across the different measures of HRV. Sometimes less effects were found for some frequency ranges resulting in quarters instead of semi circles.

Page 8: I think the authors could make it more clear that one of the conditions they were testing was the ECG component of the EEG data (extracted by ICA then projected back into the scalp space for the temporal response function analysis).

As suggested by the reviewer we adjusted our wording and replaced the arguably a bit ambiguous “... projected back separately” with “... projected back into the sensor space”. We thank the reviewer for this recommendation, as it does indeed make it easier to understand the procedure.

“After pre-processing (see Methods) the data was split in three conditions using an ICA(22). Independent components that were correlated (at $r > 0.4$; see Methods: MEG/EEG Processing - pre-processing) with the ECG electrode were either not removed from the data (Figure 3ABCD - blue), removed from the data (Figure 2ABCD - orange) or projected back into the sensor space (Figure 3ABCD - green).”

Figure 4A: standardized beta coefficients for the relationship between age and spectral slope could be noted to provide improved clarity (if I'm correct in assuming that is what they reflect).

This was indeed shown in Figure 4A and noted in the color bar as “average beta (standardized)”. We do not specifically highlight this in the text, because the exact coefficients would depend on both on the analyzed frequency range and the selected electrodes.

Figure 4I: The regressions explained at this point seems to contain a very large number of potential predictors, as I'm assuming it includes all sensors for both the ECG component and ECG rejected conditions? (if that is not the case, it could be explained in greater detail). I'm also not sure about the logic of taking a complete signal, decomposing it with ICA to separate out the ECG and non-ECG signals, then including them back into the same regression model. It seems that there could be some circularity or redundancy in doing so. However, I'm not confident that this is an issue, so would appreciate the authors explaining why it this is a valid approach (if that is the case).

After observing significant effects both in the MEG_{ECG} component and MEG_{ECG} rejected conditions in similar frequency bands we wanted to understand whether or not these age-related changes are statistically independent. To test this we added both variables as predictors in a regression model (thereby accounting for the influence of the other in relation to age). The regression models we performed were therefore actually not very complex. They were built using only two predictors, namely the data (in a specific frequency range) averaged over channels on which we noticed significant effects in the ECG rejected and ECG components data respectively (Wilkinson notation: $age \sim 1 + ECG\ rejected + ECG\ components$). This was also described in the results section stating that: “To see if MEG_{ECG} rejected and MEG_{ECG} component explain unique variance in aging at frequency ranges where we noticed shared effects, we averaged the spectral slope across significant channels and calculated a multiple regression model with MEG_{ECG} component and MEG_{ECG} rejected as predictors for age (to statistically control for the effect of MEG_{ECG} component^s and MEG_{ECG} rejected on age). This analysis was performed to understand whether the observed shared age-related effects (MEG_{ECG} rejected and MEG_{ECG} component) are in(dependent).”

We hope this explanation solves the previous misunderstanding.

The explanation of results for relationships between spectral slopes and aging reported in Figure 4 refers to clusters of effects, but the statistical inference methods section doesn't explain how these clusters were determined.

The wording of “cluster” was used to describe a “category” of effects e.g. null effects. We changed the wording from “cluster” to “category” to make this clearer stating now that: “This analysis, which is depicted in Figure 4, shows that over a broad amount of individual fitting ranges and sensors, aging resulted in a steepening of spectral slopes across conditions (see Figure 4E) with “steepening effects” observed in 25% of the processing options in MEG_{ECG} not rejected, 0.5% in MEG_{ECG} rejected, and 60% for MEG_{ECG} components. The second largest category of effects were “null effects” in 13% of the options for MEG_{ECG} not rejected, 30% in MEG_{ECG} rejected, and 7% for MEG_{ECG} components.”

Page 12: can the authors clarify whether these age related steepenings of the spectral slope in the MEG are when the data include the ECG contribution, or when the data exclude the ECG? (clarifying this seems critical to the message the authors are presenting).

We apologize for not making this clearer. We now write: “This analysis also indicates that a vast majority of observed effects irrespective of condition (ECG components, ECG not rejected, ECG rejected) show a steepening of the spectral slope with age across sensors and frequency ranges.”

Page 13: I think it would be useful to describe how much variance was explained by the MEG-ECG rejected vs MEG-ECG component conditions for a range of these analyses, so the reader also has an understanding of how much aperiodic neural activity might be influenced by age (vs if the effects are really driven mostly by changes in the ECG).

With regards to the explained variance I think that the very important question of how strong age influences changes in aperiodic activity is a topic better suited for a meta analysis. As the effect sizes seems to vary largely depending on the sample e.g. for EEG in the literature results were reported at $r=-0.08$ (Cesnaite et al. 2023), $r=-0.26$ (Cellier et al. 2021), $r=-0.24/r=-0.28/r=-0.35$ (Hill et al. 2022) and $r=0.5/r=0.7$ (Voytek et al. 2015). I would defer the reader/reviewer to the standardized beta coefficients as a measure of effect size in the current study that is depicted in Figure 4A.

Cellier, D., Riddle, J., Petersen, I., & Hwang, K. (2021). The development of theta and alpha neural oscillations from ages 3 to 24 years. *Developmental cognitive neuroscience*, 50, 100969.

Cesnaite, E., Steinfath, P., Idaji, M. J., Stephani, T., Kumral, D., Haufe, S., ... & Nikulin, V. V. (2023). Alterations in rhythmic and non-rhythmic resting-state EEG activity and their link to cognition in older age. *NeuroImage*, 268, 119810.

Hill, A. T., Clark, G. M., Bigelow, F. J., Lum, J. A., & Enticott, P. G. (2022). Periodic and aperiodic neural activity displays age-dependent changes across early-to-middle childhood. *Developmental Cognitive Neuroscience*, 54, 101076.

Voytek, B., Kramer, M. A., Case, J., Lepage, K. Q., Tempesta, Z. R., Knight, R. T., & Gazzaley, A. (2015). Age-related changes in 1/f neural electrophysiological noise. *Journal of Neuroscience*, 35(38), 13257-13265.

Also, if there are specific M/EEG sensors where the 1/f activity does relate strongly to age, it would be worth noting these, so future research could explore those sensors in more detail.

I think it is difficult to make a clear claim about this for MEG data, as the exact location or type of the sensor may differ across manufacturers. Such a statement could be easier made for source projected data or in case EEG electrodes were available, where the location would be normed eg. according to the 10-20 system.

DISCUSSION:

Page 15: Please change the wording of the following sentence, as the way it is currently worded seems to suggest that the authors of the current manuscript have demonstrated this point (which I think is not the case): "The authors demonstrate that EEG typically integrates activity over larger volumes than MEG, resulting in differently shaped spectra across both recording methods."

Apologies for the oversight! The reviewer is correct we in fact did not show this, but the authors of the cited manuscript. We correct the sentence as suggested stating now that:

“Bénar et al. demonstrate that EEG typically integrates activity over larger volumes than MEG, resulting in differently shaped spectra across both recording methods.”

Page 16: The authors mention the results can be sensitive to the application of SSS to clean the MEG data, but not ICA. I think it would be sensitive to the application of either SSS or ICA?

This is correct and actually also supported by Figure S7, as differences in ICA thresholds affect also the detection of age-related effects. We therefore adjusted the related sentences stating now that:

“ In case of the MEG signal this may include the application of Signal-Space-Separation algorithms (SSS(24,55)), different thresholds for ICA component detection (see Figure S7), high and low pass filtering, choices during spectral density estimation (window length/type etc.), different parametrization algorithms (e.g. IRASA vs FOOOF) and selection of frequency ranges for the aperiodic slope estimation.”

It would be worth clarifying that the linked mastoid re-reference alone has been proposed to cancel out the ECG signal, rather than that a linked-mastoid re-reference improves the performance of the ICA separation (which could be inferred by the explanation as it's currently written).

This is correct and we adjusted the sentence accordingly! Stating now that:

“ Previous work(12,56) has shown that a linked mastoid reference alone was particularly effective in reducing the impact of ECG related activity on aperiodic activity measured using EEG. “

The issue of the number of EEG channels could probably just be noted as a potential limitation, as could the issue of neural activity being mixed into the ECG component (although this does pose a potential confound to the M/EEG without ECG condition, I suspect it wouldn't be critical).

This is indeed a very fair point as a higher amount of electrodes would probably make it easier to better isolate ECG components in the EEG, which may be the reason why the separation did not work so well in our case. However, this is ultimately an empirical question so we highlighted it in the discussion section stating that: “Difficulties in removing ECG related components from EEG signals via ICA might be attributable to various reasons such as the number of available sensors or assumptions related to the non-gaussianity of the underlying sources. Further understanding of this matter is highly important given that ICA is the most widely used procedure to separate neural from peripheral physiological sources.”

OUTLOOK:

Page 19: Although there has been a recent trend to control for 1/f activity when examining oscillatory power, recent research suggests that this should only be implemented in specific circumstances, otherwise the correction causes more of a confound than the issue does. It might be worth considering this point with regards to the final recommendation in the Outlook section: Brake, N., Duc, F., Rokos, A., Arseneau, F., Shahiri, S., Khadra, A., & Plourde, G. (2024). A neurophysiological basis for aperiodic EEG and the background spectral trend. Nature Communications, 15(1), 1514.

We want to thank the reviewer for recommending this very interesting paper! The authors of said paper present compelling evidence showing that, while peak detection above an aperiodic trend using methods like FOOOF or IRASA is a prerequisite to determine the presence of oscillatory activity, it's not necessarily straightforward to determine which detrending approach should be applied to determine the actual power of an oscillation. Furthermore, the authors suggest that wrongfully detrending may cause larger errors than

not detrending at all. We therefore added a sentence stating that: “However, whether or not periodic activity (after detection) should be detrended using approaches like FOOOF or IRASA still remains disputed, as incorrectly detrending the data may cause larger errors than not detrending at all(75).”

RECOMMENDATIONS:

Page 20: “measure and account for” seems like it's missing a word, can this be re-written so the meaning is more clear?

Done as suggested. The sentence now states: “To better disentangle physiological and neural sources of aperiodic activity, we propose the following steps to (1) measure and (2) account for physiological influences.”

I would re-phrase “doing an ICA” to “reducing cardiac artifacts using ICA” (this wording could be changed in other places also).

I do not like to describe cardiac or ocular activity as artifactual per se. This is also why I used hyphens whenever I mention the word “artifact” in association with the ECG or EOG. However, I do understand that the wording of “doing an ICA” is a bit sloppy. We therefore reworded it accordingly throughout the manuscript to e.g. “separating cardiac from neural sources using an ICA” and “separating physiological from neural sources using an ICA”.

I would additionally note that even if components are identified as unambiguously cardiac, it is still likely that neural activity is mixed in, and so either subtracting or leaving the component will both be an issue (<https://doi.org/10.1101/2024.06.06.597688>). As such, even perfect identification of whether components are cardiac or not would still mean the issue remains (and this issue is also consistent across a considerable range of component based methods). Furthermore, current methods including wavelet transforms on the ICA component still do not provide good separation of the artifact and neural activity.

This is definitely a fair point and we also highlight this in our recommendations under 3 stating that:

“However, separating physiological from neural sources using an ICA is no guarantee that peripheral physiological activity is fully removed from the cortical signal. Even more sophisticated ICA based methods that e.g. apply wavelet transforms on the ICA components may still not provide a good separation of peripheral physiological and neural activity^{76,77}. This turns the process of deciding whether or not an ICA component is e.g. either reflective of cardiac or neural activity into a challenging problem. For instance, when we only extract cardiac components using relatively high detection thresholds (e.g. $r > 0.8$), we might end up misclassifying residual cardiac activity as neural. In turn, we can't always be sure that using lower thresholds won't result in misinterpreting parts of the neural effects as cardiac. Both ways of analyzing the data can potentially result in misconceptions.”

Castellanos, N. P., & Makarov, V. A. (2006). Recovering EEG brain signals: Artifact suppression with wavelet enhanced independent component analysis. *Journal of neuroscience methods*, 158(2), 300-312.

Bailey, N. W., Hill, A. T., Godfrey, K., Perera, M. P. N., Rogasch, N. C., Fitzgibbon, B. M., & Fitzgerald, P. B. (2024). EEG is better when cleaning effectively targets artifacts. *bioRxiv*, 2024-06.

METHODS:

Pre-processing, page 24: I assume the symmetric setting of fastica was used (rather than the deflation setting), but this should be specified.

Indeed the reviewer is correct, we used the standard setting of fastICA implemented in MNE python, which is calling the FastICA implementation in sklearn that is per default using the “parallel” or symmetric algorithm to compute an ICA. We added this information to the text accordingly, stating that:

“For extracting physiological “artifacts” from the data, 50 independent components were calculated using the *fastica* algorithm(22) (implemented in MNE-Python version 1.2; with the parallel/symmetric setting; note: 50 components were selected for MEG for computational reasons for the analysis of EEG data no threshold was applied).”

Temporal response functions, page 26: can the authors please clarify whether the TRF is computed against the ECG signal for each electrode or sensory independently, or if all electrodes/sensors are included in the analysis concurrently? I'm assuming it was computed for each electrode and sensory separately, since the TRF was computed in both the forward and backwards direction (perhaps the meaning of forwards and backwards could be explained in more detail also - i.e. using the ECG to predict the EEG signal, or using the EEG signal to predict the ECG signal?).

A TRF can also be conceptualized as a multiple regression model over time lags. This means that we used all channels to compute the forward and backward models. In the case of the forward model we predicted the signal of the M/EEG channels in a multivariate regression model using the ECG electrode as predictor. In case of the backward model we predicted the ECG electrode based on the signal of all M/EEG channels. The forward model was used to depict the time window at which the ECG signal was encoded in the M/EEG recording, which appears at 0 time lags indicating volume conduction. The backward model was used to see how much information of the ECG was decodable by taking the information of all channels.

We tried to further clarify this approach in the methods section stating that:

“We calculated the same model in the forward direction (encoding model; i.e. predicting M/EEG data in a multivariate model from the ECG signal) and backward direction (decoding model; i.e. predicting the ECG signal using all M/EEG channels as predictors).”

Page 27: the ECG data was fit using a knee, but it seems the EEG and MEG data was not.

Does this different pose any potential confound to the conclusions drawn? (having said this, Figure S4 suggests perhaps a knee was tested in the M/EEG data, which should perhaps be explained in the text also).

This was indeed tested in a previous review round to ensure that our results are not dependent on the presence/absence of a knee in the data. We therefore added figure S4, but forgot to actually add a description in the text. We are sorry for this oversight and added a paragraph to S1 accordingly:

“Using FOOOF(5), we also investigated the impact of different slope fitting options (fixed vs. knee model fits) on the aperiodic age relationship (see Supplementary Figure S4). The results that we obtained from these analyses using FOOOF offer converging evidence with our main analysis using IRASA.”

Page 32: my understanding of the result reported here is that cleaning with ICA provided better sensitivity to the effects of age on 1/f activity than cleaning with SSS. Is this

accurate? I think this could also be reported in the main manuscript, as it will be useful to researchers considering how to clean their M/EEG data prior to analyzing 1/f activity.

The reviewer is correct in stating that we overall detected slightly more “significant” effects, when not additionally cleaning the data using SSS. However, I am a bit wary of recommending omitting the use of SSS maxfilter solely based on this information. It can very well be that the higher quantity of effects (when not employing SSS maxfilter) stems from other physiological sources (e.g. muscle activity) that are correlated with age and removed when applying SSS maxfiltering. I think that just conditioning the decision of whether or not maxfilter is applied based on the amount or size of effects may not be the best idea. Instead I think that the applicability of maxfilter for research questions related to aperiodic activity should be the topic of additional methodological research. We therefore now write in Text S1:

“Considering that we detected less and weaker aperiodic effects when using SSS maxfilter is it advisable to omit maxfilter, when analyzing aperiodic signals? We don’t think that we can make such a judgment based on our current results. This is because it’s unclear whether or not the reduction of effects stems from an additional removal of peripheral information (e.g. muscle activity; that may be correlated with aging) or is induced by the SSS maxfiltering procedure itself. As the use of maxfilter in detecting changes of aperiodic activity was not subject of analysis that we are aware of, we suggest that this should be the topic of additional methodological research.”

Page 39, Figure S6 and Figure S8: Perhaps the caption could also briefly explain the difference between maxfilter set to false vs true? I might have missed it, but I didn’t gain an understanding of what varying maxfilter would mean.

Figure S6 shows the effect of ageing on the spectral slope averaged across all channels. The maxfilter set to false in AB) means that no maxfiltering using SSS was performed vs. in CD) where the data was additionally processed using the SSS maxfilter algorithm. We now describe this more clearly by writing in the caption:

“Supplementary Figure S6: Age-related changes in aperiodic brain activity are most prominent on explained by cardiac components irrespective of maxfiltering the data using signal space separation (SSS) or not AC) Age was used to predict the spectral slope (fitted at 0.1-145Hz) averaged across sensors at rest in three different conditions (ECG components not rejected [blue], ECG components rejected [orange], ECG components only [green].”

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